

Highlights

pFedSpec: Personalized Federated Learning with Spectral Allocation and Adaptive Rank Control

- KL-guided spectral allocation personalizes low-rank adaptation.
- Reinforcement learning dynamically adjusts client adapter ranks.
- pFedSpec adapts to time-varying client data distributions.
- pFedSpec outperforms baselines across text, vision, and multimodal tasks.

pFedSpec: Personalized Federated Learning with Spectral Allocation and Adaptive Rank Control

ARTICLE INFO

Keywords:

Federated Learning
Parameter-Efficient Fine-Tuning
Low-Rank Adaptation
Singular Value Decomposition
Reinforcement Learning

ABSTRACT

Large pre-trained models have demonstrated remarkable capabilities, yet adapting them in personalized federated learning remains challenging due to communication overhead, client heterogeneity, and data distribution shifts over time. To address these issues, we propose **pFedSpec**, a personalized spectral allocation framework that leverages KL-divergence to quantify client heterogeneity and dynamically assigns clients to different regions of the singular value spectrum for efficient low-rank adaptation. This allocation enables near-IID clients to exploit principal spectral components while highly non-IID clients adapt complementary subspaces, improving robustness and personalization. In addition, pFedSpec incorporates adaptive spectrum reallocation and reinforcement learning-based rank selection to accommodate time-varying data distributions. Extensive experiments on text, image, and multimodal benchmarks with DistilBERT, Qwen-0.5B, ResNet18, and CLIP ViT-B/32 demonstrate consistent superiority over state-of-the-art PEFT and personalized FL methods. Under highly heterogeneous settings, pFedSpec improves F1-score by up to 13.3%, accelerates convergence by up to 2.9 $\times$, and reduces communication cost by up to 2 $\times$ compared with existing spectral PEFT approaches. These results highlight the effectiveness of spectrum-aware adaptation for scalable and personalized federated fine-tuning of large models.

1. Introduction

Large-scale models with hundreds of millions to billions of parameters have become a dominant paradigm in modern intelligent systems, driving substantial advances in natural language processing, computer vision, and multimodal learning (Chowdhery et al., 2023; Hoffmann et al., 2022; Achiam et al., 2023; Alayrac et al., 2022; Team et al., 2023). In parallel, federated learning (FL) has emerged as a framework for decentralized training on edge devices without centralizing raw data (McMahan et al., 2017). In practice, however, deploying large-scale models in FL remains challenging due to statistical heterogeneity across clients and limited computation and communication resources (Li et al., 2020). Consequently, achieving effective personalization while maintaining adaptation efficiency remains a fundamental challenge in FL.

To address these constraints, recent studies have increasingly adopted parameter-efficient fine-tuning (PEFT) techniques, which adapt only a small subset of model parameters instead of updating the entire network (Han et al., 2024). Representative approaches include adapter-based methods (Houlsby et al., 2019), low-rank adaptation techniques such as LoRA (Hu et al., 2021), and singular value decomposition (SVD) - based decomposition methods (Meng et al., 2024; Wang et al., 2025). Several works have successfully incorporated these techniques into federated settings. For example, FedSVD (Lee et al., 2026) employs a global SVD reparameterization of LoRA to improve training stability, FFA-LoRA (Sun et al., 2024) reduces communication by freezing one adapter matrix, and FedSA-LoRA (Guo et al., 2025) separates shared and personalized knowledge through selective parameter sharing.

To further improve adaptation flexibility under heterogeneous environments, recent studies have explored rank-adaptive methods such as HetLoRA (Cho et al., 2024), LoRA-A2 (Koo et al., 2025), FlexLoRA (Bai et al., 2024), and FediLoRA (Yang et al., 2025). These approaches allow each client to adjust its parameter budget by changing the rank of low-rank adapters, enabling better alignment with system constraints and training dynamics. Despite this flexibility, rank adaptation functions primarily as a mechanism for controlling model capacity. Increasing the rank introduces more trainable parameters, while decreasing it enforces stronger compression. In existing methods, rank selection is typically selected by predefined rules, optimization heuristics, or resource-aware strategies. These approaches do not explicitly leverage client-level performance metrics—such as accuracy, F1 score, or loss—together with model size to guide rank adaptation during training, which is very important in FL environments. In heterogeneous federated settings, measuring client performance provides a direct signal of whether the current model rank is sufficient to capture local data characteristics. If performance remains low, it may indicate that the current rank is too limited to represent the underlying patterns, suggesting the need for higher capacity. Conversely, if performance saturates, increasing the rank may introduce unnecessary parameters without meaningful gains, leading to inefficiency or overfitting (Zhang et al., 2023). Therefore, using client-level performance to adjust rank enables a more informed trade-off between adaptation quality and model cost, allowing capacity to be allocated where it is most needed.

The importance of adaptive rank selection becomes even more evident in practical federated environments, where client characteristics are both non-independent and identically distributed (non-IID) and time-varying. Data distributions may drift due to changes in user behavior, environmental conditions, or application contexts, while system

ORCID(s):

resources such as computation capability, communication bandwidth, and participation frequency may also fluctuate throughout training (Gama et al., 2014; Lu et al., 2018). Under such time-varying conditions, the adaptation requirements of individual clients may evolve over time, implying that a rank configuration that is effective at one stage of training may become suboptimal later. Consequently, effective personalization requires adaptation mechanisms that can continuously respond to evolving client characteristics rather than relying on static assumptions about model capacity or client behavior.

Despite existing works in PEFT-based FL and rank-adaptive methods, a fundamental limitation remains: **clients construct their low-rank adapters using the same position in the SVD spectrum, or through the same low-rank initialization in LoRA-based approaches**. In contrast, pFedSpec not only adjusts the rank, but also assigns each client a distinct, heterogeneity-dependent spectral region for adaptation. This leads to what we term **spectral uniformity**, where adaptation is restricted to a globally shared set of representation directions regardless of client-specific data characteristics. Such a constraint is particularly limiting in non-IID and time-varying federated environments, where different clients require distinct spectral sensitivities and where the importance of spectral components may evolve over time.

Motivated by this observation, we argue that effective personalization in FL should jointly consider three complementary factors: (i) the **spectral position**, which determines which regions of the representation spectrum are most relevant to construct the adapters for a client; (ii) the **number of selected spectrum**, which determines how many spectrum contribute to the personalized adapter; and (iii) the **adaptation rank**, which control the amount of model capacity allocated to represent such information. This perspective enables personalization to adapt not only where and how broadly spectral information is exploited, but also how much adaptation capacity is required under heterogeneous and time-varying federated environments. Therefore, we propose **pFedSpec**, a personalized FL framework that enables dynamic specialization of large models in time-varying environments. pFedSpec uses a lightweight Kullback–Leibler (KL) (Goldberger et al., 2003) divergence to quantify client heterogeneity in a privacy-preserving manner, which is then used to guide spectral region assignment over SVD-decomposed model weights. Based on the heterogeneity level of each client, the top- k most relevant spectral regions are selected and combined to construct personalized low-rank adapters. Furthermore, the effective rank of these adapters is dynamically adjusted via a reinforcement learning (RL) strategy, enabling joint optimization of spectral specialization and adaptation capacity over time.

In summary, pFedSpec makes the following contributions:

- **Spectral FL personalization:** We introduce a spectral decomposition perspective for federated adaptation, enabling fine-grained control over which feature-space directions each client can update.
- **Heterogeneity-guided allocation:** We leverage KL-divergence as a privacy-preserving signal to map client heterogeneity to spectral component selection.
- **Dynamic capacity adaptation:** We incorporate a lightweight Q-learning strategy to adjust effective rank over time, enabling adaptive model complexity under evolving environments.
- **Unified framework:** We integrate spectral region assignment and adaptive rank selection into a single coherent framework, enabling dynamic and time-evolving personalization that improves both efficiency and adaptability in heterogeneous federated environments.

2. Related Work

Table 1 provides an overview of representative methods related to PEFT, personalized federated learning (pFL), and adaptive low-rank optimization. The comparison highlights several key characteristics relevant to pFedSpec, including federated deployment, personalization capability, LoRA/SVD parameterization, rank adaptation, spectral allocation strategies, and support for dynamic environments.

PiSSA (Meng et al., 2024) which applies SVD to each weight matrix and initializes the LoRA adapter matrices with its top singular values and vectors. MiLoRA (Wang et al., 2025), which inverts PiSSA’s strategy by using only the minor singular components of each weight matrix to construct adapters while freezing the principal components. However, both PiSSA and MiLoRA are the centralized methods: they do not address FL or personalization. They use a fixed-rank decomposition throughout training and assumes a static data environment. Other works extend LoRA to federated settings. FedSVD (Lee et al., 2026) also embeds SVD into federated LoRA. FedSVD applies periodic server-side SVD to orthogonalize the global adapter. Concretely, after aggregating the clients’ LoRA updates, the server factorizes the aggregate via SVD and reinitializes the adapter matrices accordingly.

On the rank-adaptive side, HetLoRA (Cho et al., 2024) introduces heterogeneous rank allocation, allowing clients with different computational capabilities to train LoRA adapters of different ranks within a unified federated optimization process. Motivated by the observation that fixed rank assignments may be suboptimal, LoRA-A2 (Koo et al., 2025) further enables adaptive rank adjustment through alternating optimization and rank selection during training. Extending adaptive rank allocation to more heterogeneous settings, FediLoRA (Yang et al., 2025) incorporates weighted aggregation strategies to effectively combine client adapters with different ranks and modality characteristics.

Table 1

Comparison of pFedSpec with related work.

Method	FL	Personalized	LoRA/SVD	Rank	Spectral Allocation	Dynamic Env
PiSSA (Meng et al., 2024)	No	No	SVD	Fixed	Principal SV(Singular Value)	Static
MiLoRA (Wang et al., 2025)	No	No	SVD	Fixed	Minor SV	Static
FedSVD (Lee et al., 2026)	Yes	No	SVD	Fixed	Principal SV	Static
HetLoRA (Cho et al., 2024)	Yes	Yes	LoRA	Adaptive	–	Static
LoRA-A2 (Koo et al., 2025)	Yes	Yes	LoRA	Adaptive	–	Static
FediLoRA (Yang et al., 2025)	Yes	Yes	LoRA	Adaptive	–	Static
FlexLoRA (Bai et al., 2024)	Yes	Yes	LoRA+SVD	Adaptive	Principal SV	Static
FedDPA (Yang et al., 2024)	Yes	Yes	LoRA	Fixed	–	Adaptive
PF2LoRA (Hao et al., 2025)	Yes	Yes	LoRA	Learned	–	Static
pFedSpec (Ours)	Yes	Yes	SVD	Adaptive	KL-guided	Adaptive

Along a similar direction, FlexLoRA (Bai et al., 2024) leverages server-side SVD reconstruction to flexibly aggregate heterogeneous LoRA adapters while preserving parameter efficiency across clients. Beyond rank adaptation, FedDPA (Yang et al., 2024) introduces a dual-path adaptation framework that combines global and local adapters through dynamic fusion to improve personalization under distribution shifts. More recently, PF2LoRA (Hao et al., 2025) advances personalized federated fine-tuning by jointly learning shared and client-specific adapters while automatically determining an appropriate rank for each client.

Overall, existing methods primarily exploit spectral information for initialization or reconstruction and largely rely on shared adaptation structures across clients. In contrast, pFedSpec enables personalized spectral allocation together with adaptive rank optimization, allowing client-specific and dynamic specialization under heterogeneous federated environments.

3. pFedSpec: Proposed method

3.1. pFedSpec Motivation

Among spectral-based PEFT approaches, PiSSA and MiLoRA are particularly relevant to our study because they explicitly exploit the SVD structure of pretrained weights, making them naturally extensible to federated environments. Their underlying motivation is further supported by SoMA (Yun et al., 2025), which shows that different singular components contribute differently to representation learning: principal singular components tend to capture dominant and generalizable features, whereas minor singular components often encode fine-grained and domain-specific information. This observation raises some important questions in personalized FL:

Research Question 1: *Why do principal singular components better serve near-IID clients, while minor singular components become more important for highly non-IID clients?*

Answer: Clients with near-IID data distributions tend to generate local updates that remain close to the globally shared optimization trajectory. As a result, their updates are naturally aligned with dominant spectral directions that

encode broadly transferable and generalizable knowledge. In contrast, highly heterogeneous clients often produce update directions that deviate from the dominant global structure due to local distribution shifts. These residual or client-specific variations may not be sufficiently represented within the principal spectral subspace and instead emerge in lower-energy spectral directions. Therefore, allocating different spectral regions according to client heterogeneity may provide a more effective personalization mechanism than enforcing a single shared spectral subspace across all clients. Empirical evidence supporting this observation is provided in Subsection 4.1, where we analyze the relationship between client heterogeneity and spectral adaptation behavior across different singular regions.

However, identifying that different clients require different spectral regions is still insufficient for practical federated environments. Even if an appropriate spectral region is assigned initially, the optimal spectral allocation for a client may change throughout training due to evolving local optimization behaviors and time-varying data distribution across communication rounds. This leads to the next question:

Research Question 2: *Why is dynamic spectrum allocation necessary instead of using fixed spectral assignment?*

Answer: In time-varying federated environments, the local data distribution of a client may change across communication rounds as new samples are added, old samples are removed, or local data continuously evolve over time in a time-series manner. A client that initially exhibits near-IID characteristics may gradually become more heterogeneous, while a highly non-IID client may later move closer to the global distribution. Consequently, the spectral region that is effective for a client in earlier communication rounds may become suboptimal later in training as the local data distribution evolves. This observation suggests that spectrum allocation should not remain fixed throughout federated optimization, but instead dynamically adapt to the changing heterogeneity characteristics of each client over time.

While dynamically adjusting the spectral region can better align personalization with evolving client heterogeneity, another important challenge still remains unresolved. Beyond determining *which position* spectral region should

be allocated, the model must also determine *how much* spectral capacity, i.e., the adaptation rank, each client requires during training. Since the rank directly controls the size of the allocated spectral subspace, it further determines the balance between personalization capability, parameter efficiency, and communication overhead for each client. This leads to the final question:

Research Question 3: *Why is adaptive rank control necessary, and why is Q-learning suitable for this problem?*

Answer: In FL, the optimal adaptation capacity for a client may continuously change throughout training due to evolving local distributions, optimization dynamics, convergence behaviors, and available hardware resources across heterogeneous devices. Some clients may require larger ranks to capture increasingly complex local variations, while others may benefit from smaller ranks to avoid overfitting, reduce communication overhead, and satisfy limited computational or memory constraints. Consequently, using a fixed rank throughout training may lead to either insufficient personalization capability or unnecessary parameter redundancy. Moreover, rank adjustment in FL is inherently sequential and dependent on long-term optimization behavior. Increasing or reducing the rank at one communication round may influence future convergence trajectories, personalization quality, and communication efficiency across subsequent rounds. Therefore, rank selection cannot be effectively modeled as a one-time static optimization problem.

To address this challenge, pFedSpec formulates rank adjustment as a sequential decision-making problem and adopts Q-learning to dynamically control client-specific rank allocation. By continuously observing training states and reward feedback across communication rounds, Q-learning enables the model to learn adaptive rank adjustment policies under non-stationary federated environments. Unlike more complex reinforcement learning approaches such as Deep Q-Networks (DQN), Q-learning provides a simple, lightweight and computationally efficient solution that is better suited for federated settings with limited communication and hardware resources. Moreover, since the rank search space is relatively small and discrete, the corresponding state-action space remains manageable without requiring deep function approximation. This allows Q-learning to effectively learn rank adaptation policies while reducing additional optimization complexity and computational overhead during federated training.

3.2. Framework Overview

Figure 1 presents the overall architecture of pFedSpec, a personalized FL framework that jointly performs heterogeneity-aware spectrum allocation and adaptive rank control.

The server first performs SVD on selected layers of the global pretrained model to construct a global spectral space. Each singular component represents a different spectral direction ordered by its relative importance, where lower spectral indices correspond to dominant and generalizable representations, while higher spectral indices capture finer-grained and client-specific variations.

To estimate client heterogeneity, each client computes the KL divergence between its local data distribution and the global distribution. The resulting KL value is then used as a lightweight heterogeneity indicator to determine the starting spectral position for each client within the global spectral space. Clients with smaller KL divergence are allocated spectral regions closer to dominant singular components, whereas highly heterogeneous clients are assigned more specialized spectral regions associated with lower-energy spectral directions.

Based on the assigned spectral position, the server constructs a personalized spectrum pool for each client using a continuous spectral window. The server then selects the top- k relevant spectral components within the allocated region to reconstruct the corresponding low-rank adapters (A_i, B_i) for local training. Detailed explanations of the window size, window positioning, and Top- K_l selection strategy are provided in Subsection 3.3.

To further handle dynamic federated environments, pFedSpec introduces a Q-learning-based rank selection mechanism that explicitly learns the next adaptation rank for each client. After local training, the client evaluates the model under the selected next-rank configuration and uses local performance, loss, and communication cost as reward signals to update the Q-table. The learned policy then determines whether the adaptation rank should be increased, decreased, or maintained in subsequent communication rounds. Since the rank directly determines the size of the allocated spectral subspace, adaptive rank control enables pFedSpec to dynamically balance personalization capability, parameter efficiency, and hardware constraints across heterogeneous client devices.

During federated optimization, only the personalized low-rank adapters are transmitted between clients and the server, while the pretrained backbone model remains frozen. After local training, the server aggregates uploaded adapters and reconstructs new personalized adapters according to the updated spectral allocation and rank decisions for the next communication round. Through this iterative process, pFedSpec continuously adapts both the spectral position and optimal rank of each client throughout training under time-varying federated environments.

3.3. pFedSpec Algorithm

Table 2 summarizes the key notations used in this section, while Algorithm 1 provides an overview of the overall pFedSpec procedure.

Client Initialization and Adapter Construction - Lines 1–5

Before federated training begins, each client $i \in K$ initializes its Q-table Q_i^1 , initial rank r_i^1 , KL divergence $D_{KL}(i)$, ϵ -greedy policy $\pi_{\epsilon,1}$, and loads the frozen pretrained model W^1 . The Q-table is defined over the rank state space $\mathcal{S} = \{2^0, 2^1, \dots, R_l\}$ and action space $\mathcal{A} = \{-R_l, \dots, -2^0, 0, 2^0, \dots, R_l\}$. Each entry $Q_i(s, a)$ estimates the expected reward of taking action a when the current rank is s . The ϵ -greedy policy selects the action with the highest

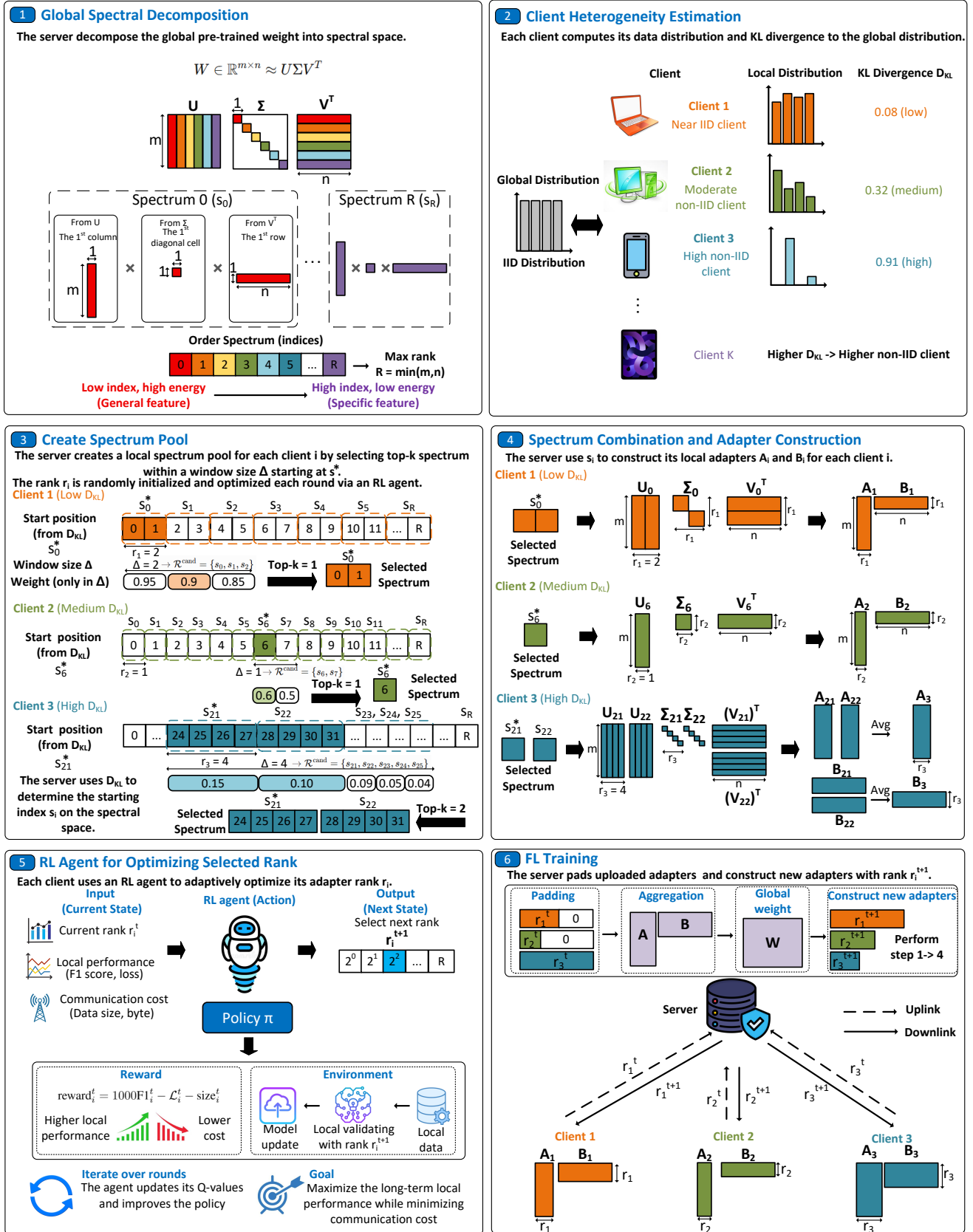


Figure 1: Overview of pFedSpec architecture.

Table 2
Notation used in pFedSpec

Symbol	Description
Data and Distribution	
K	Total number of clients.
E	Total number of epochs.
C	Total number of classes.
c	Class index, $c \in \{1, \dots, C\}$.
$n_i(c)$	Number of samples belonging to class c at client i .
N_i	Total number of samples at client i .
$\rho \in (0, 1]$	Client participation ratio.
$P_i(c)$	Probability distribution of class c in client i 's local data.
$Q(c)$	Reference (global or ideal IID) uniform distribution of class c .
$D_{\text{KL}}(P_i \ Q)$	KL divergence between $P_i(c)$ and $Q(c)$.
$KL_{\text{norm}}(i)$	Normalized KL divergence of client i .
$D_{\text{KL}}^{\min}, D_{\text{KL}}^{\max}$	Minimum and maximum KL divergence across all clients.
Model and Training	
$W^1(W_i^1)$	Frozen pretrained backbone model and its layer l , used throughout training.
$W^t, W_i^t, W_i^{t,l}$	Global model at round t , its layer l , and client i 's local model at round t .
T	Number of communication rounds.
τ	Optional update interval.
η	Learning rate.
$\mathcal{L}_i^t, \text{F1}_i^t, \text{size}_i^t$	Local loss, F1-score, model size of client i at round t .
Reinforcement Learning	
S, s	Set of possible rank states and the current rank state $s \in S$.
$\mathcal{A}, a$	Set of rank adjustment actions and the selected action $a \in \mathcal{A}$.
$Q_i^t(s, a)$	Q-table of client i for state-action pair (s, a) at round t .
π_{ϵ^t}	ϵ -greedy policy at round t .
ϵ^t	Exploration rate on round t .
$\epsilon_{\min}, \epsilon_{\max}$	Minimum and maximum exploration rates.
λ	Exponential decay coefficient for ϵ_t .
α, γ	Learning rate and discount factor in Q-learning.
r_i^t	Current selected rank of client i at round t .
Low-Rank Adapters	
R_l	Maximum rank of layer l , $R_l = \min(m, n)$.
$r_{\max}^t$	Maximum rank among clients at round t .
$A_i^t(A_{i,l}^t), B_i^t(B_{i,l}^t)$	Adapter A,B of client i at round t (and its layer l), respectively.
$\hat{A}_i^t(\hat{A}_{i,l}^t), \hat{B}_i^t(\hat{B}_{i,l}^t)$	Trained adapter A,B of client i at round t (and its layer l), respectively.
Spectrum Representation	
S_l	Number of spectral blocks at layer l .
$\text{start}_{l,s}, \text{end}_{l,s}$	Start and end indices of spectral block s at layer l .
Δ_l	Window size of the candidate spectrum at layer l .
s_l^*	Start position of the candidate spectrum at layer l .
$\mathcal{R}_l^{\text{cand}}$	Candidate spectrum set for layer l .
$\mathcal{E}_{l,s}$	Spectral energy of spectrum block s at layer l .
$w_{l,s}$	Softmax weight assigned to spectrum block s at layer l .
$ \mathcal{K}_l $	Number of selected spectrum for layer l .
$\mathcal{K}_l$	Selected Top- $ \mathcal{K}_l $ spectrum set for layer l .

Q-value for exploitation or randomly selects an action for exploration:

$$\pi_{\epsilon^t}(s) = \begin{cases} \arg \max_{a \in \mathcal{A}} Q_i(s, a), & \text{with probability } 1 - \epsilon, \\ \text{random } a \in \mathcal{A}, & \text{with probability } \epsilon. \end{cases} \quad (1)$$

Each client then sends its initial rank r_i^1 and KL divergence $D_{\text{KL}}(i)$ to the server. The KL divergence measures the statistical heterogeneity of client i by quantifying the deviation of its local class distribution $P_i(c) = \frac{n_i(c)}{N_i}$ from the uniform IID reference distribution $Q(c) = 1/C$:

$$D_{\text{KL}}(P_i \| Q) = \sum_{c=1}^C P_i(c) \log \frac{P_i(c)}{Q(c)} \quad (2)$$

A larger $D_{\text{KL}}(i)$ indicates that the client's local class distribution is more strongly skewed relative to the IID reference,

and therefore represents a higher degree of statistical heterogeneity. Since the KL values may differ substantially across clients, the server normalizes the received divergences to obtain a relative heterogeneity score $KL_{\text{norm}}(i) \in [0, 1]$. This normalized score provides a common scale for comparing clients and is subsequently used to guide the spectral allocation of their personalized adapters. Specifically, the server constructs the initial personalized adapter (A_i^1, B_i^1) for each client by invoking BUILDPERSONALIZEDADAPTERS with W^1 , r_i^1 , and $KL_{\text{norm}}(i)$ as inputs, where W^1 is the frozen pretrained model used for SVD decomposition, $KL_{\text{norm}}(i)$ determines *where* the adapter is constructed within the spectrum, and r_i^1 determines *how many* components are used to construct it. Consequently, the server generates a client-specific adapter (A_i^1, B_i^1) that reflects both the client's statistical heterogeneity and its initial rank requirement. The resulting personalized adapters are then distributed to all clients for the first communication round.

Algorithm 1 pFedSpec algorithm.

```

1: for each client  $i \in K$  do
2:   Initialize  $Q_i^1, r_i^1, D_{KL}(i), \pi_{\epsilon^1}$  and loads frozen pretrained model  $W^1$ .
3:   Send the initial rank  $r_i^1$  and KL divergence  $D_{KL}(i)$  to the server.
4: end for
5: Server computes  $KL_{\text{norm}}(i)$  and constructs the initial personalized adapters for each client.

    $(A_i^1, B_i^1) \leftarrow \text{BUILDPERSONALIZEDADAPTERS}(W^1, r_i^1, KL_{\text{norm}}(i))$  ▷ Algorithm 3

6: for each round  $t = 1$  to  $T$  do
7:   if  $t \bmod \tau = 0$  then ▷ KL-update condition
8:     Server requests updated  $D_{KL}(i)$  from all clients  $i \in K$ .
9:     Each client recomputes and sends  $D_{KL}(i)$  to the server.
10:    Server recomputes  $KL_{\text{norm}}(i)$  for all clients  $i \in K$ .
11:   end if
12:   Server samples participating clients set:  $z \leftarrow \max(\lfloor K * \rho \rfloor, 1)$ .
13:   for each client  $i \in z$  do
14:     Server sends the current personalized adapters  $(A_i^t, B_i^t)$  to client  $i$ .
15:     Client  $i$  executes  $\text{PFEDSPECCLIENT}(A_i^t, B_i^t)$  and sends  $(\hat{A}_i^t, \hat{B}_i^t)$ , and the next-round rank  $r_i^{t+1}$  to the server. ▷ Algorithm 2
16:   end for
17:   Determine  $r_{\max}^t \leftarrow \max_{i \in z}(r_i^t)$  from the current ranks of the clients and pad  $(\hat{A}_i^t, \hat{B}_i^t)$  to rank  $r_{\max}^t$ .
18:   for each selected layer  $l$  in parallel do
19:     Aggregate adapters:

       
$$A_l^{t+1} \leftarrow \sum_{i \in z} \frac{n_i}{N} \hat{A}_{i,l}^t, \quad B_l^{t+1} \leftarrow \sum_{i \in z} \frac{n_i}{N} \hat{B}_{i,l}^t.$$


20:     Update global model:  $W_l^{t+1} \leftarrow W_l^1 + A_l^{t+1}(B_l^{t+1})^\top$ .
21:   end for
22:   for each client  $i \in z$  do
23:     Server constructs the next-round personalized adapters for each client  $i$ :

       
$$(A_i^{t+1}, B_i^{t+1}) \leftarrow \text{BUILDPERSONALIZEDADAPTERS}(W^{t+1}, r_i^{t+1}, KL_{\text{norm}}(i))$$


24:   end for
25:   for each client  $i \notin z$  do ▷ Preserve the states of non-participating clients.
26:      $(A_i^{t+1}, B_i^{t+1}) \leftarrow (A_i^t, B_i^t)$ .
27:      $r_i^{t+1} \leftarrow r_i^t$ .
28:   end for
29: end for

```

Algorithm 2 pFedSpecClient algorithm.

```

Function  $\text{PFEDSPECCLIENT}(A_i^t, B_i^t)$  :
1: Validate the personalized model using  $W^1 + A_i^t(B_i^t)^\top$  to obtain  $(F1_i^t, \mathcal{L}_i^t, \text{size}_i^t)$ .
2: Compute  $\text{reward}_i^t = 1000F1_i^t - \mathcal{L}_i^t - \text{size}_i^t$ .
3: Update Q-Table:  $Q_i^{t+1}(s, a) \leftarrow Q_i^t(s, a) + \alpha [\text{reward}_i^t + \gamma \max_{a'} Q_i^t(s', a') - Q_i^t(s, a)]$ .
4: for each epoch  $e = 1$  to  $E$  ▷ Local training
5:   Train:  $\hat{A}_i^t \leftarrow A_i^t - \eta \nabla_{A_i^t} \mathcal{L}_i^t, \quad \hat{B}_i^t \leftarrow B_i^t - \eta \nabla_{B_i^t} \mathcal{L}_i^t$ .
6:   Inject:  $W_i^t \leftarrow W^1 + \hat{A}_i^t(\hat{B}_i^t)^\top$ .
7: end for
8: Update  $\epsilon^t \leftarrow \epsilon_{\min} + (\epsilon_{\max} - \epsilon_{\min})e^{-\lambda t}$ .
9: Select the next rank  $r_i^{t+1} \leftarrow \pi_{\epsilon^t}(r_i^t)$ .
10: return  $(\hat{A}_i^t, \hat{B}_i^t), r_i^{t+1}$ 

```

Federated Round and Client Participation - Lines 6–12 During federated training, the server performs T communication rounds. By default, the KL divergence computed during initialization is kept fixed, assuming that the local data distributions do not change during training. For

dynamic data environments, pFedSpec optionally supports KL-based reallocation. When this option is enabled, every τ communication rounds, each client recomputes its local class distribution and $D_{KL}(i)$ after data reallocation and

sends the updated value to the server. The server then re-computes $KL_{\text{norm}}(i)$ for all clients. In this way, the spectral allocation can be updated according to the latest data distribution while the normal federated training process remains unchanged when reallocation is not required.

At each communication round t , the server randomly selects a subset of clients according to the participation ratio ρ . The number of participating clients is determined by

$$z = \max(\lfloor K * \rho \rfloor, 1), \quad (3)$$

where K is the total number of clients and $\rho \in (0, 1]$ represents the fraction of clients selected for the current round. This partial participation reduces the computation and communication required in each round while allowing different clients to participate at different times.

Client-Side RL Agent and Adaptive Rank Selection - Lines 13–16 and Algorithm 2

For each selected client $i \in z$, the client executes the pFEDSPECCLIENT function to perform local model validation, Q-table update, local training, and adaptive rank selection. The function first validates the personalized model $W^1 + A_i^t(B_i^t)^\top$. This validation measures how well the client's current adapter configuration performs before a new rank is selected. The client obtains its local F1-score $F1_i^t$, training loss $\mathcal{L}_i^t$, and adapter communication size size_i^t . These quantities are combined into a single reward:

$$\text{reward}_i^t = 1000F1_i^t - \mathcal{L}_i^t - \text{size}_i^t \quad (4)$$

The reward is designed to capture the three optimization objectives considered by pFedSpec. The F1-score measures predictive performance and is therefore positively rewarded, while the training loss and adapter size are penalized because they represent lower optimization quality and higher communication overhead, respectively. The factor 1000 scales the F1-score contribution so that it has a sufficiently strong influence relative to the loss and communication terms. Consequently, a rank configuration receives a higher reward when it provides better predictive performance without unnecessarily increasing training loss or communication cost.

The obtained reward is then used to update the client's Q-table according to the Q-learning rule:

$$Q_i^{t+1}(s, a) \leftarrow Q_i^t(s, a) + \alpha \left[\text{reward}_i^t + \gamma \max_{a'} Q_i^t(s', a') - Q_i^t(s, a) \right]. \quad (5)$$

Here, $Q_i^t(s, a)$ represents the current Q-value of taking action a when the current adapter rank is represented by state s . The immediate reward reward_i^t tells the client how useful the current rank configuration was, while $\max_{a'} Q_i^t(s', a')$ estimates the best future utility available after reaching the resulting state s' . The learning rate α controls how strongly the newly obtained experience changes the existing Q-value, whereas the discount factor γ determines how much future rewards are considered. Through repeated updates, the Q-table estimates the long-term utility of each rank-adjustment

action, enabling the policy to balance model performance against resource cost.

After updating the Q-table, the client performs E local training epochs. During local training, only the personalized adapter parameters are optimized, while the pretrained backbone remains frozen:

$$\hat{A}_i^t \leftarrow A_i^t - \eta \nabla_{A_i^t} \mathcal{L}_i^t, \quad \hat{B}_i^t \leftarrow B_i^t - \eta \nabla_{B_i^t} \mathcal{L}_i^t, \quad (6)$$

where η denotes the local learning rate. This parameter update follows the PEFT principle because the large pretrained backbone is not updated. Instead, the client only optimizes the compact low-rank matrices, reducing the number of trainable parameters and the amount of information that needs to be communicated.

After each local update, the trained adapters are combined with the frozen backbone to obtain the locally adapted model:

$$W_i^t \leftarrow W^1 + \hat{A}_i^t(\hat{B}_i^t)^\top. \quad (7)$$

This operation injects the client-specific low-rank update into the pretrained model. Thus, the local model retains the knowledge encoded in W^1 while incorporating information learned from the client's private data through $(\hat{A}_i^t, \hat{B}_i^t)$.

After completing local training, the client updates its exploration probability according to the exponentially decaying schedule:

$$\epsilon^t = \epsilon_{\min} + (\epsilon_{\max} - \epsilon_{\min})e^{-\lambda t}, \quad (8)$$

where $\epsilon_{\max}$ and $\epsilon_{\min}$ define the initial and minimum exploration probabilities, respectively, and λ controls the decay rate. The purpose of this schedule is to gradually change the behavior of the RL agent from exploration to exploitation. At early communication rounds, a relatively large ϵ^t allows the client to try different rank-adjustment actions and collect diverse experience. As training progresses, ϵ^t decreases, causing the client to increasingly rely on the actions with higher learned Q-values.

Using the updated exploration probability, the client selects the rank for the next communication round according to

$$r_i^{t+1} \leftarrow \pi_{\epsilon^t}(r_i^t). \quad (9)$$

Here, the current rank r_i^t represents the state of the RL agent, while the selected action determines how the rank is changed for the next round. The selected r_i^{t+1} therefore represents the new adaptation capacity that the client requests for its next personalized adapter. After selecting the next rank, the client returns the locally trained adapters $(\hat{A}_i^t, \hat{B}_i^t)$ together with r_i^{t+1} to the server.

Server Aggregation - Line 17–21

After receiving the locally updated adapters $(\hat{A}_i^t, \hat{B}_i^t)$ and the selected next ranks r_i^{t+1} from the participating clients, the server determines the maximum rank of the current adapters based on their current ranks r_i^t :

$$r_{\max}^t = \max_{i \in z}(r_i^t). \quad (10)$$

This operation is necessary because different clients may use different adapter ranks. Consequently, their received adapter matrices may have different dimensions. The server uses $r_{\max}^t$ as a common rank so that all received adapters can be represented in the same parameter space. The adapters with smaller ranks are padded to this common dimension before aggregation. Padding allows the server to aggregate adapters of different original ranks while preserving their existing learned components.

For each selected layer l , the server then aggregates the padded adapters using the relative sizes of the participating clients' local datasets:

$$A_l^{t+1} \leftarrow \sum_{i \in \mathcal{Z}} \frac{n_i}{N} \hat{A}_{i,l}^t, \quad B_l^{t+1} \leftarrow \sum_{i \in \mathcal{Z}} \frac{n_i}{N} \hat{B}_{i,l}^t, \quad (11)$$

where n_i denotes the number of training samples at client i and $N = \sum_{i \in \mathcal{Z}} n_i$ denotes the total number of training samples across the participating clients. The factor n_i/N gives a larger aggregation weight to clients with more local training data. This weighted aggregation combines the local adapter updates into a shared global adapter while accounting for differences in the amount of training data contributed by each client.

The aggregated adapters are then combined with the frozen pretrained backbone to obtain the updated global model:

$$W_l^{t+1} \leftarrow W_l^1 + A_l^{t+1}(B_l^{t+1})^\top. \quad (12)$$

Here, W_l^1 denotes the corresponding layer of the frozen pretrained backbone, while A_l^{t+1} and B_l^{t+1} contain the knowledge aggregated from the participating clients. The resulting W^{t+1} therefore provides an updated shared model that incorporates the information learned during the current communication round. This updated model is subsequently used as the input for the next personalized spectral allocation.

Personalized Adapters Reconstruction - Lines 22–24

For each participating client i , the server constructs a new personalized adapter by invoking BUILDPERSONALIZEDADAPTERS with the updated global model W^{t+1} , the selected rank r_i^{t+1} , and the normalized KL score $KL_{\text{norm}}(i)$ as inputs. The detailed procedure for this spectrum-guided adapter construction is described in Algorithm 3.

State Preservation for Non-Participating Clients - Lines 25–28

For clients that do not participate in the current round, no local training or rank selection is performed. Their personalized states are therefore preserved:

$$(A_i^{t+1}, B_i^{t+1}) \leftarrow (A_i^t, B_i^t), \quad r_i^{t+1} \leftarrow r_i^t. \quad (13)$$

This means that a non-participating client keeps the latest personalized adapter and rank configuration instead of being replaced by the aggregated global adapter. When the client is selected in a future communication round, its preserved personalized state is used as the starting point for the next local update. In this way, pFedSpec maintains client-specific personalization even under partial client participation.

Spectrum-Guided Personalized Adapter Construction - Algorithm 3

While Algorithm 1 describes the overall federated optimization workflow of pFedSpec, Algorithm 3 specifies how the BUILDPERSONALIZEDADAPTERS function constructs a personalized adapter from the latest global model. Specifically, the function uses the latest global model W^t together with the latest available $KL_{\text{norm}}(i)$ to construct a new personalized adapter with the selected rank r_i^t for each client i .

For each selected layer $l \in W^t$, the server first performs SVD on the current global model:

$$W_l^t = U_l \Sigma_l V_l^\top, \quad (14)$$

where U_l and V_l contain the left and right singular vectors, respectively, and Σ_l contains the corresponding singular values. The singular values are ordered from larger to smaller values, allowing the layer to be viewed as an ordered spectrum of components with different levels of spectral energy. Instead of always selecting the principal singular components, pFedSpec divides this spectrum into contiguous blocks according to the target rank r_i^t . Each block contains up to r_i^t consecutive singular components, so that a selected block can directly provide the components needed to construct a rank- r_i^t adapter. The number of available blocks is determined by $S_l = \max(1, \lfloor R_l/r_i^t \rfloor)$, where $R_l = \min(m, n)$ is the maximum rank of layer l .

After dividing the spectrum into blocks, pFedSpec uses the client's normalized KL divergence to select a suitable region of the spectrum. Instead of searching all spectrum blocks, the server first defines a candidate window. Its size and starting position are determined by the client's target rank and normalized KL divergence:

$$\Delta_l \leftarrow \min(r_i^t, S_l), s_l^* \leftarrow \lfloor KL_{\text{norm}}(i)(S_l - \Delta_l) \rfloor. \quad (15)$$

Here, Δ_l specifies how many spectrum blocks are considered, while s_l^* specifies where the candidate window starts. The candidate spectrum set is then defined as

$$\mathcal{R}_l^{\text{cand}} \leftarrow \{s \mid s_l^* \leq s < s_l^* + \Delta_l\}. \quad (16)$$

In simple terms, $KL_{\text{norm}}(i)$ determines where the candidate window is placed: a smaller value places it closer to the beginning of the spectrum, while a larger value moves it toward later spectrum blocks. Thus, each client is given a different candidate spectrum region according to its statistical heterogeneity. The server then evaluates the spectrum blocks within $\mathcal{R}_l^{\text{cand}}$ to identify the most informative ones.

After obtaining the candidate spectrum set $\mathcal{R}_l^{\text{cand}}$, pFedSpec evaluates the importance of each candidate spectrum block based on its spectral energy. For each candidate block s , the server calculates

$$\mathcal{E}_{l,s} \leftarrow \|\Sigma_l[\text{start}_{l,s} : \text{end}_{l,s}]\|^2, \forall s \in \mathcal{R}_l^{\text{cand}}. \quad (17)$$

The spectral energy measures the strength of the singular components contained in a spectrum block. A larger $\mathcal{E}_{l,s}$ means that the block contains larger singular values

Algorithm 3 Spectrum-Guided Personalized Adapter Construction.

Function BUILDPERSONALIZEDADAPTERS($W^t, r_i^t, KL_{\text{norm}}(i)$)

- 1: **for** each selected layer l in W^t **do**
- 2: Perform SVD: $W_l^t = U_l \Sigma_l V_l^\top$.
- 3: Number of spectrum: $S_l \leftarrow \max(1, \lfloor R_l / r_i^t \rfloor)$, where $R_l = \min(m, n)$.
- 4: Calculate $\text{start}_{l,s} \leftarrow s \cdot r_i^t$, $\text{end}_{l,s} \leftarrow \min((s+1) \cdot r_i^t, R_l)$, $\forall s \in \{0, \dots, S_l - 1\}$.
- 5: Determine candidate spectrum window:

$$\Delta_l \leftarrow \min(r_i^t, S_l), s_l^* \leftarrow \lfloor KL_{\text{norm}}(i)(S_l - \Delta_l) \rfloor, \mathcal{R}_l^{\text{cand}} \leftarrow \{s \mid s_l^* \leq s < s_l^* + \Delta_l\}.$$
- 6: Spectral energy:

$$\mathcal{E}_{l,s} \leftarrow \|\Sigma_l[\text{start}_{l,s} : \text{end}_{l,s}]\|^2, \quad \forall s \in \mathcal{R}_l^{\text{cand}}.$$
- 7: Softmax:

$$w_{l,s} \leftarrow \frac{\exp(\mathcal{E}_{l,s} - \max_j \mathcal{E}_{l,j})}{\sum_{j \in \mathcal{R}_l^{\text{cand}}} \exp(\mathcal{E}_{l,j} - \max_j \mathcal{E}_{l,j})}, \quad \forall s \in \mathcal{R}_l^{\text{cand}}.$$
- 8: Select Top- $\mathcal{K}$ spectrum:

$$|\mathcal{K}_l| \leftarrow \min\left(\max\left(1, \left\lfloor \frac{|\mathcal{R}_l^{\text{cand}}|}{2} \right\rfloor\right), |\mathcal{R}_l^{\text{cand}}|\right),$$

$$\mathcal{K}_l \leftarrow \text{indices of the top } |\mathcal{K}_l| \text{ weights among } \{w_{l,s} \mid s \in \mathcal{R}_l^{\text{cand}}\}.$$
- 9: Construct spectrum adapters:

$$A_{l,s}^t \leftarrow U_l[:, \text{start}_{l,s} : \text{end}_{l,s}] \Sigma_l[\text{start}_{l,s} : \text{end}_{l,s}]^{1/2},$$

$$B_{l,s}^t \leftarrow \Sigma_l[\text{start}_{l,s} : \text{end}_{l,s}]^{1/2} V_l[\text{start}_{l,s} : \text{end}_{l,s} :]^\top, \quad \forall s \in \mathcal{R}_l^{\text{cand}}.$$
- 10: Combine adapters:

$$A_{l,i}^t = \frac{1}{|\mathcal{K}_l|} \sum_{s \in \mathcal{K}_l} w_{l,s} A_{l,s}^t, \quad B_{l,i}^t = \frac{1}{|\mathcal{K}_l|} \sum_{s \in \mathcal{K}_l} w_{l,s} B_{l,s}^t.$$
- 11: **end for**
- 12: Assemble personalized adapters:

$$A_i^t \leftarrow \{A_{l,i}^t\}, \quad B_i^t \leftarrow \{B_{l,i}^t\}.$$
- 13: **return** (A_i^t, B_i^t).

and therefore represents more information from the current model. This allows the server to distinguish the more informative blocks within the candidate region rather than treating all of them equally.

The spectral energies are then converted into normalized importance weights using a softmax function:

$$w_{l,s} \leftarrow \frac{\exp(\mathcal{E}_{l,s} - \max_j \mathcal{E}_{l,j})}{\sum_{j \in \mathcal{R}_l^{\text{cand}}} \exp(\mathcal{E}_{l,j} - \max_j \mathcal{E}_{l,j})}, \forall s \in \mathcal{R}_l^{\text{cand}}. \quad (18)$$

These weights indicate the relative importance of the candidate spectrum blocks. A block with higher spectral energy

receives a larger weight and therefore has a stronger influence on the final personalized adapter. The subtraction of the maximum energy is used only for numerical stability when computing the exponential values.

Based on these importance weights, pFedSpec applies Top- $\mathcal{K}$ selection to retain only the most informative blocks within the candidate spectrum. The number of selected blocks is determined by

$$|\mathcal{K}_l| \leftarrow \min\left(\max\left(1, \left\lfloor \frac{|\mathcal{R}_l^{\text{cand}}|}{2} \right\rfloor\right), |\mathcal{R}_l^{\text{cand}}|\right). \quad (19)$$

This rule selects approximately half of the candidate blocks. The inner $\max(\cdot)$ ensures that at least one block is selected, while the outer $\min(\cdot)$ ensures that the number of selected blocks never exceeds the number of available candidate blocks.

The motivation for introducing the Top- $\mathcal{K}$ selection is to allow pFedSpec to utilize multiple informative spectrum blocks within the candidate set $\mathcal{R}_l^{\text{cand}}$, particularly when the target adapter rank is sufficiently large. For clients with small ranks, such as $r_i^t \in 1, 2$, the candidate set $\mathcal{R}_l^{\text{cand}}$ covers only a small portion of the full SVD spectrum. In this case, selecting multiple blocks in $\mathcal{K}_l$ provides limited additional diversity because the selected blocks are located within a narrow spectral region and may contain similar or highly correlated information. In contrast, when the target rank r_i^t is larger, the candidate set $\mathcal{R}_l^{\text{cand}}$ covers a broader spectral region. The blocks within this larger region can capture more diverse directions of variation, making the selection of multiple informative blocks in $\mathcal{K}_l$ more beneficial. Therefore, Top- $\mathcal{K}$ allows pFedSpec to exploit richer spectral information when the available candidate region is sufficiently broad, while naturally limiting the number of selected blocks when the candidate region is small.

The selected spectrum blocks are then determined from their importance weights. Thus, $\mathcal{K}_l$ contains the most informative spectrum blocks within the candidate set $\mathcal{R}_l^{\text{cand}}$. For each selected block $s \in \mathcal{K}_l$, pFedSpec constructs a rank- r_i^t candidate adapter using the corresponding singular vectors and singular values:

$$\begin{aligned} A_{l,s}^t &\leftarrow U_l[:, \text{start}_{l,s} : \text{end}_{l,s}] \Sigma_l[\text{start}_{l,s} : \text{end}_{l,s}]^{1/2}, \\ B_{l,s}^t &\leftarrow \Sigma_l[\text{start}_{l,s} : \text{end}_{l,s}]^{1/2} V_l[\text{start}_{l,s} : \text{end}_{l,s}, :]^T, \end{aligned} \quad (20)$$

When multiple spectrum blocks are selected, pFedSpec combines their candidate adapters using the corresponding importance weights:

$$\begin{aligned} A_{i,l}^t &= \frac{1}{|\mathcal{K}_l|} \sum_{s \in \mathcal{K}_l} w_{l,s} A_{l,s}^t, \\ B_{i,l}^t &= \frac{1}{|\mathcal{K}_l|} \sum_{s \in \mathcal{K}_l} w_{l,s} B_{l,s}^t. \end{aligned} \quad (21)$$

The weights $w_{l,s}$ determine how much each selected spectrum block contributes to the final adapter. A block with a larger weight has a larger contribution, while a block with a smaller weight has a smaller contribution. The selected blocks are not concatenated. Instead, they are first used to construct multiple candidate adapters, each with the same target rank r_i^t , and these candidate adapters are then combined by weighted averaging. Therefore, even when multiple spectrum blocks are selected, the final personalized adapter still has rank r_i^t and its communication cost does not increase. The resulting layer-wise adapters are then combined across the selected layers to form the complete personalized adapter (A_i^t, B_i^t) for client i . In this way, Top- $\mathcal{K}$ allows the final adapter to use information from multiple informative spectrum blocks while keeping the same adapter size and communication overhead.

4. Theoretical Analysis

We analyze the optimization effect of structural spectrum under a fixed client partition. The purpose of this section is not to prove convergence of every dynamic component of the full training system, but to isolate the mathematical benefit of assigning statistically similar clients to the same spectral group.

4.1. Effect of Spectral Component Selection on Reconstruction Quality

Figure. 2 provides an overall visualization of the pre-trained weight matrix, its SVD decomposition, and the reconstructions obtained from the principal and minor spectral components. Figure. 2a shows the original pretrained weight matrix W before spectral decomposition. Let the pretrained weight matrix be decomposed as:

$$W = U \Sigma V^T = \sum_{i=1}^r \sigma_i u_i v_i^T, \quad (22)$$

where $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r$ are the singular values, and u_i and v_i denote the corresponding left and right singular vectors. Figure. 2b visualizes the three factors of the SVD decomposition, namely the left singular-vector matrix U , the diagonal singular-value matrix Σ , and the right singular-vector matrix V^T . The red and green regions indicate the principal and minor spectral components, respectively. In particular, the principal components correspond to the top- k terms associated with the largest singular values, whereas the minor components correspond to the tail of the spectrum associated with substantially smaller singular values. Based on this spectral partitioning, the corresponding components can be reconstructed separately to examine how different regions of the spectrum contribute to the original parameter structure.

As shown in Figure. 2c, the reconstruction using principal components,

$$W_{\text{principal}} = \sum_{i=1}^k \sigma_i u_i v_i^T, \quad (23)$$

closely matches the original weight matrix in Figure. 2a, indicating that most of the Frobenius energy

$$\|W\|_F^2 = \sum_{i=1}^r \sigma_i^2 \quad (24)$$

is concentrated in the dominant singular directions. In contrast, the minor component reconstruction,

$$W_{\text{minor}} = \sum_{i=k+1}^r \sigma_i u_i v_i^T, \quad (25)$$

shown in Figure. 2d, exhibits significantly lower magnitude (approximately within the range $[0, 0.04]$) and captures only residual high-frequency perturbations, with limited structural alignment to the original parameter geometry.

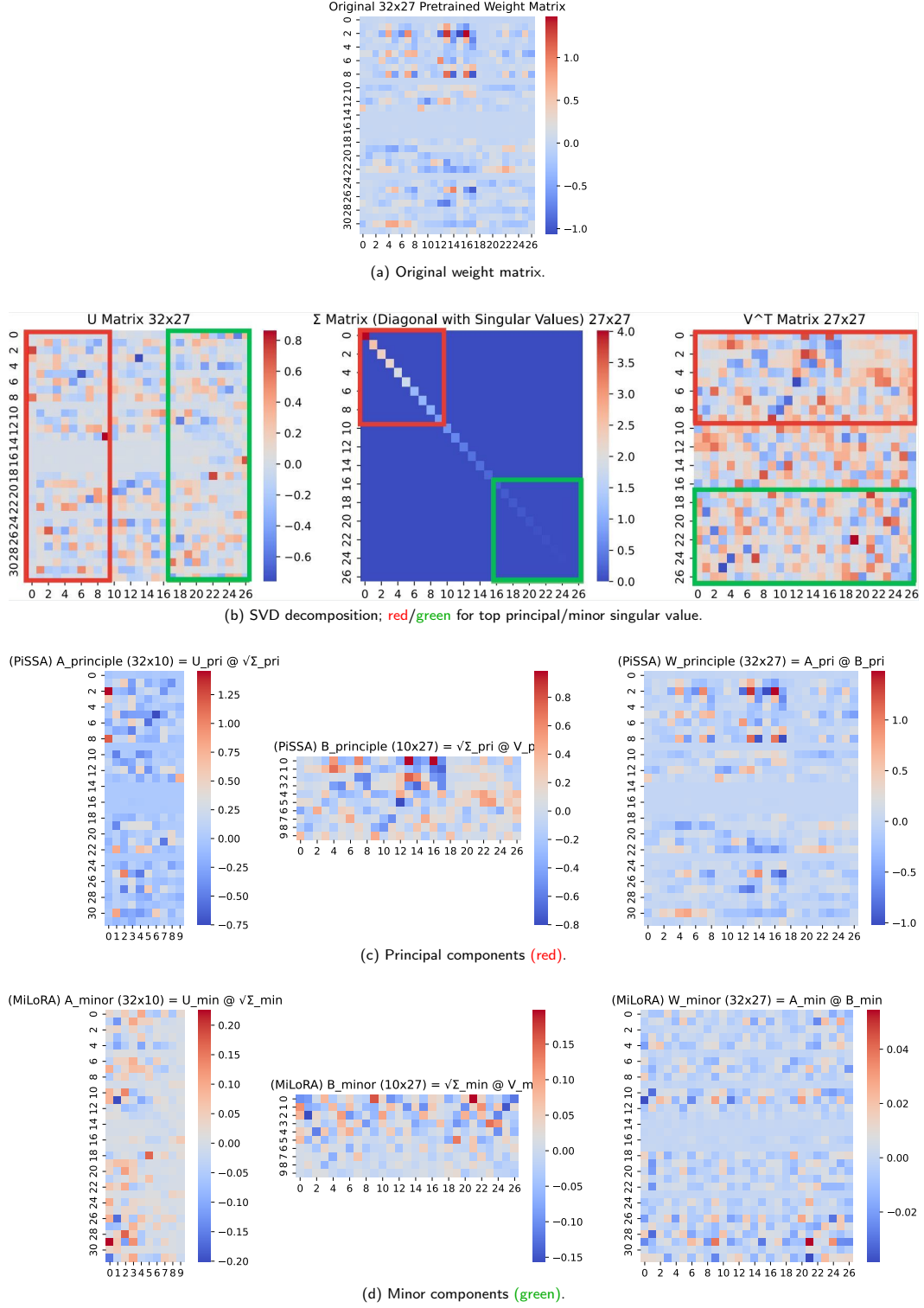


Figure 2: Matrix visualizations of pretrained weights and their low-rank decompositions.

Key insight. Since pFedSpec constructs client adapters as linear combinations of spectral slices derived from this decomposition, the contribution of each component to the effective update is proportional to its singular energy σ_i^2 .

Principal components therefore dominate the induced update direction:

$$\Delta W_i \propto \sum_{j \in S_i} w_j \sigma_j u_j v_j^T, \quad (26)$$

where S_i denotes the selected spectral subset for client i . In contrast, adapters constructed from minor components yield updates with substantially smaller norm, as $\sigma_j \rightarrow 0$ in the tail spectrum.

As a consequence, clients assigned to lower-energy spectral regions—typically associated with higher heterogeneity and more specialized structures—produce updates that have limited impact on the global aggregation. Meanwhile, updates aligned with principal spectral directions are reinforced during averaging due to their higher magnitude and stability across clients.

Overall, these observations empirically validate the spectral bias underlying pFedSpec aggregation: combining adapters across spectral regions does not introduce destructive interference, but instead preserves dominant representation directions while attenuating low-energy, highly localized perturbations. This explains why spectral decomposition-based aggregation remains stable and effective even under heterogeneous client participation.

4.2. Convergence Under Structural Spectrum

Let $F_k(w)$ denote the local objective of client $k \in [K]$, and let

$$F(w) = \frac{1}{K} \sum_{k=1}^K F_k(w) \quad (27)$$

be the global objective. Define

$$g_k(w) := \nabla F_k(w), \quad g(w) := \frac{1}{K} \sum_{k=1}^K g_k(w). \quad (28)$$

Assume that the spectrum mechanism induces a fixed partition

$$C = \{C_1, \dots, C_M\} \quad (29)$$

of the clients during the analysis horizon. For each group C_m , define the group mean gradient

$$\bar{g}_m(w) := \frac{1}{|C_m|} \sum_{k \in C_m} g_k(w). \quad (30)$$

We consider two heterogeneity measures. The first is the standard global gradient heterogeneity:

$$\zeta^2(w) := \frac{1}{K} \sum_{k=1}^K \|g_k(w) - g(w)\|^2. \quad (31)$$

The second is the within group heterogeneity induced by spectrum:

$$\zeta_{\text{in}}^2(w; C) := \frac{1}{K} \sum_{m=1}^M \sum_{k \in C_m} \|g_k(w) - \bar{g}_m(w)\|^2. \quad (32)$$

Assumption 1 Each local objective F_k is L smooth.

Assumption 2 Each client has an unbiased stochastic gradient oracle. That is, for any iterate w ,

$$\mathbb{E}[\hat{g}_k(w)] = g_k(w), \quad \mathbb{E}\|\hat{g}_k(w) - g_k(w)\|^2 \leq \sigma^2. \quad (33)$$

Proposition 1 For any partition C ,

$$\zeta^2(w) = \zeta_{\text{in}}^2(w; C) + \zeta_{\text{btw}}^2(w; C), \quad (34)$$

where

$$\zeta_{\text{btw}}^2(w; C) := \frac{1}{K} \sum_{m=1}^M |C_m| \|\bar{g}_m(w) - g(w)\|^2. \quad (35)$$

Proof sketch Use the identity

$$g_k(w) - g(w) = (g_k(w) - \bar{g}_m(w)) + (\bar{g}_m(w) - g(w)), \quad k \in C_m, \quad (36)$$

expand the squared norm, and sum over all clients. The cross term vanishes because

$$\sum_{k \in C_m} (g_k(w) - \bar{g}_m(w)) = 0 \quad (37)$$

for every group C_m .

Corollary 1 For any partition C ,

$$\zeta_{\text{in}}^2(w; C) \leq \zeta^2(w), \quad (38)$$

with equality if and only if $\bar{g}_m(w) = g(w)$ for all m .

Interpretation Proposition 1 gives an exact variance decomposition. Structural spectrum does not increase the heterogeneity term seen inside each spectral group. When clients with similar objectives are grouped together, the effective optimization difficulty is therefore reduced.

We next state a standard nonconvex stationarity bound under fixed spectrum.

Assumption 3 The client partition C remains fixed over the analyzed communication horizon of length T .

Theorem 1 Under Assumptions 1 to 3, consider local SGD with a sufficiently small stepsize η . Then there exist constants $c_1, c_2 > 0$, depending only on standard optimization parameters such as the number of local steps and participation pattern, such that

$$\frac{1}{T} \sum_{t=1}^{T-1} \mathbb{E}\|\nabla F(w_t)\|^2 \leq \frac{2(F(W^1) - F_{\text{inf}})}{\eta T} + c_1 \eta \sigma^2 + c_2 \eta \bar{\zeta}_{\text{in}}^2, \quad (39)$$

where

$$\bar{\zeta}_{\text{in}}^2 := \frac{1}{T} \sum_{t=0}^{T-1} \zeta_{\text{in}}^2(w_t; C), \quad (40)$$

and F_{inf} is a lower bound of F .

Remark Relative to a uniform adaptation analysis, the heterogeneity dependent term is governed by within group variation rather than global variation. By Corollary 1, this term is never larger than the original global heterogeneity term. Hence, any client partition that reduces ζ_{in}^2 yields a strictly tighter stationarity bound.

4.3. Justification of KL Guided Spectral Allocation

We now justify KL divergence as a mathematically meaningful routing criterion. Let P_k denote the data distribution of client k , and let R_m denote the reference distribution associated with spectral component m . Define

$$F_k(w) := \mathbb{E}_{z \sim P_k}[\ell(w; z)], \quad G_m(w) := \mathbb{E}_{z \sim R_m}[\ell(w; z)]. \quad (41)$$

The objective of routing is to assign each client to a spectral component whose reference distribution is close to the client distribution. We show that KL closeness controls gradient mismatch.

Assumption 4 For every sample z and parameter vector w , the per-sample gradient is bounded:

$$\|\nabla \ell(w; z)\| \leq G. \quad (42)$$

Assumption 5 Whenever $\text{KL}(P_k \| R_m)$ is evaluated, P_k is absolutely continuous with respect to R_m .

Proposition 2 Under Assumptions 4 and 5, for any client k and spectral component m ,

$$\|\nabla F_k(w) - \nabla G_m(w)\| \leq 2G \text{TV}(P_k, R_m) \leq G \sqrt{2 \text{KL}(P_k \| R_m)}. \quad (43)$$

Consequently,

$$\|\nabla F_k(w) - \nabla G_m(w)\|^2 \leq 2G^2 \text{KL}(P_k \| R_m). \quad (44)$$

Proof sketch Write

$$\nabla F_k(w) - \nabla G_m(w) = \int \nabla \ell(w; z) (dP_k(z) - dR_m(z)). \quad (45)$$

Taking norms and using $\|\nabla \ell(w; z)\| \leq G$ gives

$$\|\nabla F_k(w) - \nabla G_m(w)\| \leq G \int |dP_k - dR_m| = 2G \text{TV}(P_k, R_m). \quad (46)$$

Applying Pinsker's inequality,

$$\text{TV}(P_k, R_m) \leq \sqrt{\text{KL}(P_k \| R_m)/2}, \quad (47)$$

yields the result.

KL routing objective We define the routing objective:

$$\mathcal{J}(\phi) := \frac{1}{K} \sum_{k=1}^K \text{KL}(P_k \| R_{\phi(k)}), \quad (48)$$

where $\phi : [K] \rightarrow [M]$ maps each client to a spectral component. We also define the average gradient mismatch:

$$\Delta(\phi; w) := \frac{1}{K} \sum_{k=1}^K \|\nabla F_k(w) - \nabla G_{\phi(k)}(w)\|^2. \quad (49)$$

Corollary 2 For any routing map ϕ ,

$$\Delta(\phi; w) \leq 2G^2 \mathcal{J}(\phi). \quad (50)$$

Interpretation Corollary 2 shows that minimizing KL divergence between client and spectral-component distributions minimizes an upper bound on gradient mismatch. Therefore, KL-based routing is not a heuristic similarity score but a principled surrogate for reducing optimization conflict.

Remark (on normalization) In pFedSpec, KL values used for spectral scoring are computed from empirical distributions and may be rescaled for numerical stability during implementation. However, the theoretical analysis is carried out on the standard KL divergence $\text{KL}(P_k \| R_m)$ without normalization. Introducing a normalized variant is unnecessary in the theoretical setting, since all bounds rely on the fundamental properties of KL divergence and Pinsker's inequality.

Proposition 3 Let

$$\phi^* \in \arg \min_{\phi} \mathcal{J}(\phi). \quad (51)$$

If a routing rule $\hat{\phi}$ satisfies

$$\mathcal{J}(\hat{\phi}) \leq \mathcal{J}(\phi^*) + \varepsilon, \quad (52)$$

then for every w ,

$$\Delta(\hat{\phi}; w) \leq 2G^2 (\mathcal{J}(\phi^*) + \varepsilon). \quad (53)$$

Interpretation Proposition 3 implies graceful degradation. Even if routing is only approximately optimal in KL, the resulting gradient mismatch remains controlled up to an additive error proportional to ε .

Summary This analysis provides two complementary insights. First, spectrum reduces global heterogeneity into structured subspace assignments, tightening optimization behavior. Second, KL divergence provides a theoretically grounded routing signal that upper bounds client-spectral-component gradient mismatch, thereby supporting coherent spectral allocation in heterogeneous FL.

5. Experiments

5.1. Experimental Setup

In this section, we present the experimental protocol used to evaluate pFedSpec. Our goal is to provide a comprehensive comparison across diverse FL settings, with particular emphasis on data heterogeneity, personalization, and communication efficiency.

5.1.1. Dataset and Data Partitioning

We evaluate pFedSpec on four benchmarks spanning multiple modalities: 20NewsGroup (Adinugroho et al., 2022) and Banking77 (Loukas et al., 2023) for text classification, CIFAR100 (Zheng et al., 2024) for image classification, and UMPC-Food101 (Bossard et al., 2014) for multimodal classification involving both visual and textual inputs. This diverse collection of datasets allows us to assess the generality of pFedSpec across different tasks, modalities, and model architectures. To simulate statistical heterogeneity, the training data are partitioned across clients using a Dirichlet distribution (Arisdakessian et al., 2025) with concentration parameter $\alpha \in 0.05, 0.01$, where smaller values correspond to more severe non-IID settings. Each client has access to its entire local dataset throughout training.

We consider two evaluation scenarios. In the *static* setting, the client data distributions remain fixed throughout all communication rounds, representing the standard FL benchmark. In the *time-varying* setting, clients continuously receive additional local samples during training, resulting in gradually evolving data distributions and changing levels of heterogeneity over time. To capture these distributional changes, KL divergence scores are periodically recomputed and used to update the client heterogeneity signals.

5.1.2. Baselines

We compare pFedSpec against a diverse set of baselines including centralized federated learning (cFL) and personalized federated learning (pFL).

For cFL methods, we consider full-finetuning (FFT), LoRA Hu et al. (2021), PiSSA Meng et al. (2024), and MiLoRA Wang et al. (2025). These methods learn a single global model shared by all clients and differ primarily in their adaptation mechanisms. FFT updates all model parameters, LoRA performs low-rank adaptation through trainable adapter matrices, while PiSSA and MiLoRA further exploit SVD by leveraging principal and minor spectral components, respectively.

For pFL methods, we include FlexLoRA Bai et al. (2024), LoRA-A2 Koo et al. (2025), and FedSVD Lee et al. (2026), which introduce client-specific adaptation through personalized parameter sharing and SVD-based federated optimization. These methods serve as strong baselines for handling statistical heterogeneity in federated environments. Unless otherwise specified, all methods are evaluated under identical data partitions, model architectures, participation rates, and communication settings to ensure a fair comparison.

5.1.3. Training Details And Implementation

Model architectures For text classification tasks, we employ DistilBERT (Sanh et al., 2019) and Qwen1.5-0.5B (Patil et al., 2026) as the backbone models. For image classification, we use ResNet18 (Liu et al., 2024), while the multimodal experiments on UMPC-Food101 are conducted using a CLIP-based architecture (Shen et al., 2021). Spectral adapters are applied to all linear layers in Transformer-based models (Dosovitskiy et al., 2021) and to the 1×1 projection layers in ResNet18.

Training configuration Unless otherwise specified, the default setting consists of 10 clients with a participation rate of 30% per communication round. All methods are trained for 1000 communication rounds with one local epoch per round and a batch size of 32. Learning rates are selected from 10^{-2} , 10^{-3} , 10^{-4} using validation performance.

Evaluation metrics We evaluate model performance using the best mean F1-score across clients and the minimum average training loss attained throughout the federated optimization process. To assess communication efficiency, we additionally report the number of communication rounds required to reach a target F1-score, total training time, and cumulative communication cost. The target F1-score is defined as the maximum performance achieved by LoRA under the most challenging heterogeneity setting. All reported results are averaged over three independent runs.

Compute Environment All experiments are conducted on a workstation equipped with four NVIDIA RTX A6000 GPUs. All methods are evaluated under the same hardware and software environment to ensure fair comparison.

5.1.4. pFedSpec Setup

To understand the contribution of each component in pFedSpec, we conduct a structured ablation under four configurations.

pFedSpec (full functionality) pFedSpec jointly performs KL-guided spectral allocation and RL-based rank adaptation, where each client is assigned a heterogeneity-dependent spectral region in the SVD space and dynamically adjusts its effective rank over communication rounds based on local performance.

pFedSpec w/o Spectral Allocation (w/o SA) Spectral allocation is removed while RL-based rank adaptation is retained, so all clients share the same SVD spectral region (principal values) and only the effective rank varies across training.

pFedSpec w/o RL RL-based rank adaptation is removed while spectral allocation is kept, so each client operates in a KL-dependent spectral region but uses a fixed set rank $r = 32$ throughout training.

Table 3

F1 Score Performance on 20News and Banking77.

Setting			$\alpha = 0.05$		$\alpha = 0.01$	
Model	Domain	Method	20News	Banking77	20News	Banking77
DistilBERT	cFL	FFT	0.8104 $\pm$ 0.003	0.9476 $\pm$ 0.004	0.8100 $\pm$ 0.006	0.9489 $\pm$ 0.002
		LoRA	0.748 $\pm$ 0.013	0.9328 $\pm$ 0.006	0.7433 $\pm$ 0.007	0.896 $\pm$ 0.001
		PiSSA	0.7969 $\pm$ 0.004	0.9442 $\pm$ 0.005	0.8023 $\pm$ 0.001	0.9455 $\pm$ 0.003
		MiLoRA	0.7583 $\pm$ 0.014	0.9390 $\pm$ 0.005	0.7579 $\pm$ 0.001	0.9157 $\pm$ 0.007
	pFL	FlexLoRA	0.782 $\pm$ 0.005	0.944 $\pm$ 0.003	0.788 $\pm$ 0.006	0.938 $\pm$ 0.005
		LoRA-A2	0.786 $\pm$ 0.004	0.945 $\pm$ 0.004	0.792 $\pm$ 0.004	0.940 $\pm$ 0.004
		FedSVD	0.792 $\pm$ 0.004	0.946 $\pm$ 0.003	0.798 $\pm$ 0.003	0.943 $\pm$ 0.004
	Ours	pFedSpec	0.872 $\pm$ 0.008	0.956 $\pm$ 0.003	0.905 $\pm$ 0.004	0.972 $\pm$ 0.005
		w/o SA	0.810 $\pm$ 0.007	0.941 $\pm$ 0.004	0.820 $\pm$ 0.005	0.955 $\pm$ 0.005
		w/o RL	0.862 $\pm$ 0.007	0.951 $\pm$ 0.004	0.892 $\pm$ 0.005	0.965 $\pm$ 0.004
		w/o SA + RL	0.796 $\pm$ 0.008	0.947 $\pm$ 0.005	0.802 $\pm$ 0.006	0.944 $\pm$ 0.005
QWEN-0.5B	cFL	FFT	0.8157 $\pm$ 0.005	0.9238 $\pm$ 0.001	0.822 $\pm$ 0.004	0.917 $\pm$ 0.008
		LoRA	0.706 $\pm$ 0.009	0.8623 $\pm$ 0.005	0.711 $\pm$ 0.002	0.854 $\pm$ 0.009
		PiSSA	0.7664 $\pm$ 0.012	0.8821 $\pm$ 0.003	0.780 $\pm$ 0.005	0.888 $\pm$ 0.006
		MiLoRA	0.7362 $\pm$ 0.013	0.8721 $\pm$ 0.013	0.751 $\pm$ 0.016	0.883 $\pm$ 0.002
	pFL	FlexLoRA	0.758 $\pm$ 0.006	0.884 $\pm$ 0.004	0.770 $\pm$ 0.006	0.889 $\pm$ 0.005
		LoRA-A2	0.762 $\pm$ 0.005	0.886 $\pm$ 0.004	0.774 $\pm$ 0.005	0.891 $\pm$ 0.005
		FedSVD	0.768 $\pm$ 0.005	0.889 $\pm$ 0.003	0.780 $\pm$ 0.004	0.894 $\pm$ 0.004
	Ours	pFedSpec	0.805 $\pm$ 0.007	0.918 $\pm$ 0.004	0.823 $\pm$ 0.006	0.921 $\pm$ 0.006
		w/o SA	0.752 $\pm$ 0.006	0.874 $\pm$ 0.004	0.794 $\pm$ 0.006	0.882 $\pm$ 0.005
		w/o RL	0.776 $\pm$ 0.006	0.897 $\pm$ 0.004	0.807 $\pm$ 0.006	0.89 $\pm$ 0.005
		w/o SA + RL	0.766 $\pm$ 0.007	0.890 $\pm$ 0.005	0.78 $\pm$ 0.007	0.884 $\pm$ 0.006

pFedSpec w/o SA + RL Both spectral allocation and RL are disabled, resulting in a static setting where all clients use the same spectral region (principal values) and a fixed low-rank configuration $r = 32$ across all communication rounds.

5.2. Main Results

This subsection presents a comprehensive evaluation of pFedSpec from multiple perspectives. We first examine its performance on text, image, and multimodal datasets to assess its effectiveness across different data modalities and model architectures. We then investigate its behavior under time-varying data distributions and analyze the relationship between client heterogeneity, KL divergence, spectral structure, and rank dynamics. Finally, we conduct an ablation study on Top- $\mathcal{K}$ spectrum selection to further examine the effect of the spectral allocation strategy.

5.2.1. Text dataset

Table 3 reports the F1-score performance on 20News and Banking77 under $\alpha = 0.05$ and $\alpha = 0.01$ using DistilBERT and QWEN-0.5B. Overall, pFedSpec consistently achieves the best performance across all settings, with the advantage becoming more pronounced under the severe heterogeneity setting $\alpha = 0.01$. For example, with DistilBERT, pFedSpec achieves 0.905 and 0.972 on 20News and Banking77, respectively, outperforming the strongest baseline, PiSSA, by 0.103 and 0.027, while with QWEN-0.5B it improves over FedSVD from 0.780 to 0.823 on

20News and from 0.894 to 0.921 on Banking77. Notably, pFedSpec maintains its performance advantage as the data become more heterogeneous, with the largest gains observed in the $\alpha = 0.01$ setting. The ablation results further show that removing spectral allocation causes a larger performance drop than removing RL-based rank adaptation in most settings, while removing both components leads to the largest degradation. For instance, under DistilBERT with $\alpha = 0.01$, removing SA reduces the F1-score from 0.905 to 0.820 on 20News, whereas removing RL reduces it to 0.892; disabling both reduces it further to 0.802. These results demonstrate that spectral allocation is the primary contributor to the performance gain, while RL-based rank adaptation provides a complementary benefit by dynamically adjusting the adaptation capacity to client requirements.

Table 4 compares the communication rounds, training time, and cumulative communication cost required to reach the target F1-score. Overall, pFedSpec reaches the target with substantially fewer rounds than the existing pFL baselines across both datasets and backbone models; for example, under $\alpha = 0.01$, it requires only 70 and 48 rounds on 20News with DistilBERT and QWEN-0.5B, compared with 195 and 130 rounds for FedSVD, respectively. The reduction in rounds is also reflected in training time and communication cost, with pFedSpec achieving particularly large savings for QWEN-0.5B, where the cost on 20News decreases from 47.50 GB with FedSVD to 17.54 GB at $\alpha = 0.01$. Compared

Table 4

Communication Efficiency on 20News and Banking77 Datasets.

Setting			Round		Time (hrs)		Cost (GB)	
Dataset	Model	Method	$\alpha = 0.05$	$\alpha = 0.01$	$\alpha = 0.05$	$\alpha = 0.01$	$\alpha = 0.05$	$\alpha = 0.01$
20News		FFT	88.33	126.33	0.321	0.565	141.97	203.05
		LoRA	819.33	795	3.067	3.026	53.09	51.51
		PiSSA	294.33	326.33	1.158	1.242	19.07	21.14
		MiLoRA	563.66	601.33	2.167	2.302	36.52	38.96
	DistilBERT (Target F1: 0.73)	FlexLoRA	217	245	0.78	0.91	13.60	15.90
		LoRA-A2	182	208	0.68	0.78	14.85	17.05
		FedSVD	175	195	0.63	0.74	11.02	12.64
		pFedSpec (Ours)	95	70	0.81	0.62	10.16	9.54
		w/o SA	175	205	0.88	1.02	16.84	18.76
		w/o RL	118	92	0.84	0.69	12.35	11.88
		w/o SA + RL	245	285	1.17	1.38	18.94	20.25
		FFT	41	41	2.167	2.159	456.59	456.59
		LoRA	796	920.66	35.67	42.77	290.85	336.41
		PiSSA	105	91.66	4.258	4.172	38.36	33.49
		MiLoRA	253	293	13.01	13.29	92.44	107.06
	QWEN-0.5B (Target F1: 0.7)	FlexLoRA	140	160	6.10	7.00	51.15	58.46
		LoRA-A2	122	138	5.30	6.05	47.85	54.65
		FedSVD	115	130	5.00	5.70	42.02	47.50
		pFedSpec (Ours)	80	48	2.20	1.75	29.23	17.54
		w/o SA	125	80	2.85	2.30	49.75	35.62
		w/o RL	95	58	2.45	1.95	36.40	24.88
		w/o SA + RL	165	105	3.55	2.85	61.85	45.93
Banking77		FFT	51	68	0.588	0.752	81.97	109.29
		LoRA	568.66	938.6	3.892	7.47	36.84	60.82
		PiSSA	270	306.3	2.251	2.444	12.96	19.85
		MiLoRA	467	704.3	4.53	5.267	30.26	43.69
	DistilBERT (Target F1: 0.89)	FlexLoRA	240	280	1.10	1.28	15.55	18.14
		LoRA-A2	215	245	0.98	1.13	17.85	20.90
		FedSVD	205	235	0.94	1.09	13.28	15.22
		pFedSpec (Ours)	150	165	0.62	0.70	19.72	21.69
		w/o SA	195	220	0.81	0.93	32.45	36.82
		w/o RL	170	185	0.69	0.78	24.88	27.56
		w/o SA + RL	235	265	0.98	1.12	39.55	43.68
		FFT	20	29.66	0.675	1.091	222.72	330.37
		LoRA	396	536.66	15.672	17.015	144.69	196.09
		PiSSA	58	78	2.241	2.40	21.19	28.50
		MiLoRA	195	234.33	6.72	7.218	71.25	85.62
	QWEN-0.5B (Target F1: 0.8)	FlexLoRA	90	120	3.20	4.10	32.88	43.85
		LoRA-A2	78	102	2.75	3.55	30.80	40.25
		FedSVD	72	95	2.55	3.20	26.32	34.71
		pFedSpec (Ours)	40	35	2.25	2.10	24.62	22.78
		w/o SA	58	50	2.85	2.65	39.80	36.95
		w/o RL	46	40	2.40	2.25	30.35	28.16
		w/o SA + RL	68	58	3.35	3.05	48.25	44.10

with FFT, pFedSpec also provides a much lower communication cost despite FFT requiring fewer rounds, highlighting the high per-round cost of full-model transmission. The ablation results further show that removing spectral allocation increases both convergence rounds and communication cost, while removing RL generally causes a smaller increase;

for example, on 20News with QWEN-0.5B at $\alpha = 0.01$, pFedSpec requires 48 rounds and 17.54 GB, compared with 80 rounds and 35.62 GB for w/o SA and 58 rounds and 24.88 GB for w/o RL. Disabling both components results in the highest cost among the pFedSpec variants, confirming

Table 5

Performance on CIFAR100 dataset.

Setting			F1 score		Round		Time (hrs)		Cost (GB)	
Dataset	Model	Method	$\alpha = 0.1$	$\alpha = 0.01$	$\alpha = 0.1$	$\alpha = 0.01$	$\alpha = 0.1$	$\alpha = 0.01$	$\alpha = 0.1$	$\alpha = 0.01$
CIFAR100	ResNet18 (Target F1: 0.5)	FFT	0.591 $\pm$ 0.053	0.583 $\pm$ 0.077	109 $\pm$ 8.11	173 $\pm$ 19.77	0.460 $\pm$ 0.030	0.488 $\pm$ 0.080	29.288	46.48
		LoRA	0.513 $\pm$ 0.051	0.501 $\pm$ 0.047	912 $\pm$ 48.11	957 $\pm$ 73.11	2.867 $\pm$ 0.970	2.955 $\pm$ 0.880	6.201	6.507
		PiSSA	0.567 $\pm$ 0.075	0.554 $\pm$ 0.134	318 $\pm$ 21.34	370 $\pm$ 25.78	1.132 $\pm$ 0.050	1.347 $\pm$ 0.090	2.162	2.516
		MiLoRA	0.546 $\pm$ 0.073	0.542 $\pm$ 0.033	566 $\pm$ 54.32	582 $\pm$ 48.96	1.782 $\pm$ 0.120	1.999 $\pm$ 0.080	3.848	3.957
		pFedEdit	0.565 $\pm$ 0.033	0.537 $\pm$ 0.022	425 $\pm$ 20.18	560 $\pm$ 18.67	1.218 $\pm$ 0.171	2.688 $\pm$ 0.221	10.85	12.57
		FedAS	0.541 $\pm$ 0.077	0.535 $\pm$ 0.087	488 $\pm$ 16.33	623 $\pm$ 25.44	2.157 $\pm$ 0.110	2.518 $\pm$ 0.150	17.94	25.34
		FlexLoRA	0.555 $\pm$ 0.030	0.548 $\pm$ 0.028	402 $\pm$ 18.20	455 $\pm$ 21.30	1.380 $\pm$ 0.090	1.610 $\pm$ 0.120	2.734	3.094
		LoRA-A2	0.558 $\pm$ 0.028	0.551 $\pm$ 0.030	388 $\pm$ 16.50	442 $\pm$ 19.40	1.322 $\pm$ 0.080	1.548 $\pm$ 0.100	3.086	3.486
		FedSVD	0.562 $\pm$ 0.027	0.556 $\pm$ 0.029	365 $\pm$ 15.70	418 $\pm$ 18.10	1.265 $\pm$ 0.070	1.472 $\pm$ 0.090	2.482	2.842
		pFedSpec (Ours)	0.601$\pm$0.010	0.612$\pm$0.012	248$\pm$12.30	221$\pm$9.85	1.342$\pm$0.060	1.201$\pm$0.040	2.686	2.503
	w/o SA w/o RL w/o SA + RL	w/o SA	0.586 $\pm$ 0.011	0.598 $\pm$ 0.013	275 $\pm$ 13.40	248 $\pm$ 10.20	1.410 $\pm$ 0.065	1.265 $\pm$ 0.045	3.864	3.548
		w/o RL	0.588 $\pm$ 0.011	0.600 $\pm$ 0.013	282 $\pm$ 13.80	255 $\pm$ 10.50	1.430 $\pm$ 0.066	1.280 $\pm$ 0.046	2.431	2.322
		w/o SA + RL	0.579 $\pm$ 0.012	0.592 $\pm$ 0.014	290 $\pm$ 14.10	265 $\pm$ 11.00	1.455 $\pm$ 0.070	1.310 $\pm$ 0.050	4.112	3.925

Table 6

Performance comparison on multimodal dataset UMPC-Food101.

Method	F1-score (target = 0.89)			Loss	Convergence		Time each round (s)	Cost	
	Last	Max	Avg		Round	Time (h)		Cost/Round (MB)	Total Cost (GB)
FFT	0.9415 $\pm$ 0.001	0.9477 $\pm$ 0.001	0.9262 $\pm$ 0.0008	10.4 $\pm$ 0.57	22.6 $\pm$ 1.24	0.84 $\pm$ 0.07	150.8 $\pm$ 0.789	7441.29	164.15
LoRA	0.8972 $\pm$ 0.008	0.9058 $\pm$ 0.004	0.6694 $\pm$ 0.03	56.419 $\pm$ 2.8	358.6 $\pm$ 6.64	14.219 $\pm$ 1.63	134.9 $\pm$ 9.508	442.33	154.73
PiSSA	0.9207 $\pm$ 0.01	0.9337 $\pm$ 0.003	0.8756 $\pm$ 0.005	39.41 $\pm$ 0.722	81.33 $\pm$ 4.642	3.226 $\pm$ 0.398	141.26 $\pm$ 0.933	442.33	35.14
MiLoRA	0.9199 $\pm$ 0.002	0.9258 $\pm$ 0.0009	0.7775 $\pm$ 0.016	49.75 $\pm$ 2.116	194 $\pm$ 14.71	7.367 $\pm$ 0.46	141.38 $\pm$ 0.59	442.33	83.74
FlexLoRA	0.925 $\pm$ 0.004	0.931 $\pm$ 0.003	0.842 $\pm$ 0.010	41.20 $\pm$ 1.80	112.50 $\pm$ 2.40	5.10 $\pm$ 0.20	140.90 $\pm$ 2.90	520.00	57.10
LoRA-A2	0.928 $\pm$ 0.004	0.934 $\pm$ 0.003	0.848 $\pm$ 0.009	39.80 $\pm$ 1.70	106.80 $\pm$ 2.20	4.82 $\pm$ 0.18	141.10 $\pm$ 2.80	505.00	53.93
FedSVD	0.932 $\pm$ 0.003	0.938 $\pm$ 0.003	0.855 $\pm$ 0.008	37.50 $\pm$ 1.60	102.40 $\pm$ 2.10	4.60 $\pm$ 0.17	141.30 $\pm$ 2.70	490.00	49.02
pFedSpec	0.952$\pm$0.003	0.955$\pm$0.003	0.888$\pm$0.006	28.10$\pm$1.50	62.40$\pm$1.10	2.65$\pm$0.05	139.80$\pm$2.40	410.00	25.02
w/o SA	0.938 $\pm$ 0.004	0.942 $\pm$ 0.004	0.872 $\pm$ 0.008	31.20 $\pm$ 1.70	70.80 $\pm$ 1.40	3.02 $\pm$ 0.06	140.10 $\pm$ 2.60	470.00	33.28
w/o RL	0.940 $\pm$ 0.004	0.943 $\pm$ 0.004	0.875 $\pm$ 0.008	30.80 $\pm$ 1.60	72.60 $\pm$ 1.50	3.08 $\pm$ 0.06	140.30 $\pm$ 2.50	610.00	44.17
w/o SA + RL	0.935 $\pm$ 0.005	0.939 $\pm$ 0.005	0.868 $\pm$ 0.009	32.40 $\pm$ 1.80	79.20 $\pm$ 1.60	3.28 $\pm$ 0.07	140.00 $\pm$ 2.70	640.00	50.69

that spectral allocation and adaptive rank selection jointly contribute to the communication efficiency of pFedSpec.

Taken together, Tables 3 and 4 demonstrate that pFedSpec provides a favorable balance between F1-score performance and communication efficiency across different datasets, backbone models, and levels of data heterogeneity. pFedSpec consistently achieves the highest F1-scores while requiring fewer communication rounds and lower cumulative communication cost than the existing pFL baselines. The advantage is particularly evident under the more challenging $\alpha = 0.01$ setting, where pFedSpec maintains higher F1-scores while reaching the target F1-score with fewer communication rounds. The ablation results further confirm that spectral allocation is the main contributor to the F1-score improvement, while RL-based rank adaptation provides complementary benefits in convergence and communication efficiency. Overall, these results demonstrate that jointly adapting the spectral location and rank of low-rank adapters enables pFedSpec to achieve stronger F1-score performance with improved communication efficiency in heterogeneous FL.

5.2.2. Image dataset

Table 5 demonstrates that pFedSpec generalizes effectively to vision tasks and consistently achieves the best performance on CIFAR100 under both heterogeneity settings. Compared with existing PEFT and personalized FL baselines, pFedSpec attains the highest F1 score while requiring substantially fewer communication rounds. It achieves the highest F1-score under both heterogeneity levels, reaching 0.601 at $\alpha = 0.1$ and 0.612 at $\alpha = 0.01$, while requiring only 248 and 221 communication rounds, respectively, to reach the target F1-score.

The ablation results further confirm the complementary roles of spectral allocation and RL-based rank adaptation. Removing either component leads to noticeable degradation in both accuracy and convergence efficiency, while disabling both components results in the worst performance among the pFedSpec variants. In particular, the larger degradation observed in the w/o SA setting suggests that spectral allocation contributes most to the performance gains, whereas RL-based rank adaptation further accelerates convergence by dynamically adjusting the adapter capacity.

Regarding communication cost, the full pFedSpec achieves a favorable trade-off between efficiency and accuracy. Although the RL exploration mechanism introduces a slightly

higher communication overhead than the fixed-rank w/o RL variant, it enables faster convergence and ultimately yields the best overall performance-cost balance. These results indicate that the proposed spectral personalization mechanism is effective not only for language models but also for vision architectures such as ResNet18.

5.2.3. Multimodal dataset

Table 6 reports the results on the multimodal UMPC-Food101 dataset. Compared with existing PEFT and personalized FL baselines, **pFedSpec** achieves the highest final, maximum, and average F1-scores while simultaneously attaining the lowest training loss. In particular, pFedSpec reaches a maximum F1-score of 0.955, outperforming the strongest baseline FedSVD (0.938), indicating that jointly exploiting spectral personalization and adaptive rank selection is also beneficial in multimodal settings.

From an efficiency perspective, pFedSpec converges in only 62.4 rounds and requires 2.65 hours of training, representing a reduction of approximately 39% in communication rounds and 42% in training time compared with FedSVD. Moreover, it incurs the lowest communication overhead among all PEFT methods, with only 25.02 GB of total communication cost, nearly half that of FedSVD (49.02 GB).

The ablation study further reveals the contributions of the two components. Removing spectral allocation (w/o SA) causes a noticeable degradation in F1-score and increases the total communication cost from 25.02 GB to 33.28 GB, suggesting that assigning heterogeneous spectral regions effectively reduces interference among clients. In contrast, removing RL-based rank adaptation (w/o RL) leads to a smaller accuracy drop but substantially increases communication cost to 44.17 GB due to the use of a fixed rank throughout training. Disabling both components (w/o SA + RL) results in the lowest performance and the highest communication overhead among the ablation variants.

Overall, these results demonstrate that spectral allocation primarily improves personalization quality, whereas RL-based rank adaptation mainly enhances communication efficiency. Their combination enables pFedSpec to achieve the best accuracy-efficiency trade-off on multimodal federated learning tasks.

5.2.4. Time-varying data

Table 7 evaluates the robustness of pFedSpec under time-varying client distributions, where local data statistics evolve periodically with intervals $\tau \in 10, 100$. Overall, pFedSpec consistently achieves the highest F1 score while maintaining low communication overhead across both datasets and heterogeneity settings.

Compared with existing PEFT methods, pFedSpec exhibits significantly stronger adaptation capability under non-stationary environments. Conventional methods such as LoRA and MiLoRA suffer noticeable performance degradation and require a large number of communication rounds to recover from distribution shifts. Although spectral approaches, including FlexLoRA, LoRA-A2, and FedSVD,

improve convergence speed, their fixed rank structures limit their ability to adapt to evolving client distributions. In contrast, pFedSpec dynamically reallocates spectral components and adjusts effective ranks according to the changing local characteristics, resulting in substantially higher accuracy and faster convergence.

The effect of the drift interval is also evident. When the distribution changes more frequently ($\tau = 10$), pFedSpec maintains strong performance by continuously updating the spectral allocation, demonstrating its capability to rapidly track local distribution changes. Increasing the interval to $\tau = 100$ leads to a slight degradation in accuracy and a moderate increase in communication rounds, since the adaptation process becomes less responsive to newly emerging patterns. Nevertheless, the performance drop remains relatively small compared with competing methods, indicating that the learned spectral representations possess strong stability and generalization properties.

Another notable observation is that pFedSpec benefits more from highly heterogeneous environments ($\alpha = 0.01$). Across both 20News and Banking77, pFedSpec achieves larger performance gains over the baselines under severe non-IID settings. This suggests that personalized spectrum allocation effectively captures client-specific information and mitigates the adverse effects of distribution mismatch among clients.

The ablation study further highlights the importance of both SA and RL. Removing either component consistently reduces F1 scores and increases convergence time, while removing both components causes the most severe degradation. Specifically, eliminating SA restricts the model's ability to adaptively select informative spectral regions under distribution shifts, whereas removing RL limits the flexibility of client-specific capacity allocation. The results indicate that the two modules complement each other and jointly contribute to maintaining robustness in dynamic federated environments.

Finally, despite operating under time-varying data distributions, pFedSpec preserves low communication cost and competitive training time. This demonstrates that the proposed adaptive spectrum mechanism improves robustness against concept drift without sacrificing communication efficiency, making pFedSpec suitable for realistic FL scenarios where client data distributions evolve over time.

5.2.5. Client Heterogeneity, KL Structure, and Rank Dynamics

To better understand pFedSpec under highly non-IID settings, we analyze the relationship between client data heterogeneity, measured via KL_{norm} , and adaptive model capacity through rank allocation. Table 8 reports per-client statistics for both 20News and Banking77 under $\alpha = 0.01$, including number of samples, number of classes, normalized KL divergence, and the initial rank assigned at the start of training.

The results show strong structural heterogeneity across clients. Clients with fewer samples and fewer observed

Table 7

Performance on 20NewsGroup and Banking77 dataset with time-varying data.

Setting			F1 score		Round		Time (hrs)		Cost (GB)	
Dataset	Model	Method	$\alpha = 0.05$	$\alpha = 0.01$	$\alpha = 0.05$	$\alpha = 0.01$	$\alpha = 0.05$	$\alpha = 0.01$	$\alpha = 0.05$	$\alpha = 0.01$
20News	Distilbert (Target F1: 0.69) (Interval τ : 10 rounds)	FFT	0.7950 $\pm$ 0.0010	0.8100 $\pm$ 0.0070	84.00 $\pm$ 20.92	91.00 $\pm$ 7.87	0.697 $\pm$ 0.167	0.732 $\pm$ 0.056	135.01	146.26
		LoRA	0.7049 $\pm$ 0.0020	0.7126 $\pm$ 0.0100	860.00 $\pm$ 48.00	877.00 $\pm$ 27.58	6.55 $\pm$ 0.404	6.85 $\pm$ 0.209	55.72	56.82
		PiSSA	0.7639 $\pm$ 0.0140	0.7823 $\pm$ 0.0030	439.00 $\pm$ 33.13	357.00 $\pm$ 39.82	3.25 $\pm$ 0.265	2.67 $\pm$ 0.286	28.44	23.13
		MiLoRA	0.7271 $\pm$ 0.0200	0.7315 $\pm$ 0.0140	752.33 $\pm$ 101.69	705.33 $\pm$ 69.14	5.70 $\pm$ 0.799	5.58 $\pm$ 0.428	48.75	45.71
		FlexLoRA	0.7817 $\pm$ 0.0048	0.7876 $\pm$ 0.0056	216.28 $\pm$ 17.72	244.12 $\pm$ 20.61	0.80 $\pm$ 0.085	0.89 $\pm$ 0.112	13.51	15.81
		LoRA-A2	0.7854 $\pm$ 0.0037	0.7913 $\pm$ 0.0039	190.44 $\pm$ 15.82	219.31 $\pm$ 17.64	0.70 $\pm$ 0.076	0.81 $\pm$ 0.087	12.22	14.11
		FedSVD	0.7914 $\pm$ 0.0036	0.7972 $\pm$ 0.0028	174.61 $\pm$ 14.52	194.21 $\pm$ 13.73	0.62 $\pm$ 0.066	0.73 $\pm$ 0.078	10.96	12.57
		pFedSpec	0.8715$\pm$0.0075	0.9038$\pm$0.0037	94.28$\pm$8.64	69.18$\pm$5.73	0.78$\pm$0.057	0.61$\pm$0.047	10.11	9.46
		w/o SA	0.8342 $\pm$ 0.0058	0.8786 $\pm$ 0.0047	238.71 $\pm$ 21.34	178.94 $\pm$ 17.23	0.93 $\pm$ 0.106	1.17 $\pm$ 0.133	12.72	10.82
		w/o RL	0.8413 $\pm$ 0.0049	0.8839 $\pm$ 0.0054	228.82 $\pm$ 19.51	174.32 $\pm$ 15.74	0.90 $\pm$ 0.097	1.15 $\pm$ 0.126	12.03	10.31
		w/o SA + RL	0.8272 $\pm$ 0.0066	0.8705 $\pm$ 0.0057	258.43 $\pm$ 23.61	193.52 $\pm$ 19.11	1.00 $\pm$ 0.115	1.25 $\pm$ 0.142	13.32	11.38
20News	Distilbert (Target F1: 0.69) (Interval τ : 100 rounds)	FFT	0.7992 $\pm$ 0.0020	0.8081 $\pm$ 0.0080	84.00 $\pm$ 25.31	74.66 $\pm$ 3.86	0.668 $\pm$ 0.195	0.647 $\pm$ 0.063	135.01	120.01
		LoRA	0.7195 $\pm$ 0.0007	0.6943 $\pm$ 0.0040	821.00 $\pm$ 87.00	884.00 $\pm$ 26.00	6.297 $\pm$ 0.723	7.320 $\pm$ 0.299	53.20	57.28
		PiSSA	0.7690 $\pm$ 0.0090	0.7806 $\pm$ 0.0070	423.30 $\pm$ 75.52	359.00 $\pm$ 25.46	2.963 $\pm$ 0.537	2.578 $\pm$ 0.191	27.43	23.26
		MiLoRA	0.7281 $\pm$ 0.0120	0.7285 $\pm$ 0.0050	720.33 $\pm$ 50.60	739.66 $\pm$ 21.79	5.15 $\pm$ 0.371	5.99 $\pm$ 0.803	46.67	47.93
		FlexLoRA	0.7896 $\pm$ 0.0058	0.7947 $\pm$ 0.0048	199.34 $\pm$ 16.72	224.18 $\pm$ 17.63	0.69 $\pm$ 0.078	0.83 $\pm$ 0.087	12.36	14.02
		LoRA-A2	0.7926 $\pm$ 0.0047	0.7985 $\pm$ 0.0039	179.42 $\pm$ 15.63	209.37 $\pm$ 16.71	0.65 $\pm$ 0.067	0.78 $\pm$ 0.077	11.63	13.71
		FedSVD	0.7954 $\pm$ 0.0038	0.8013 $\pm$ 0.0029	164.28 $\pm$ 14.57	189.11 $\pm$ 13.62	0.59 $\pm$ 0.058	0.71 $\pm$ 0.068	10.74	12.21
		pFedSpec	0.8735$\pm$0.0057	0.9072$\pm$0.0038	91.34$\pm$7.62	64.27$\pm$4.71	0.77$\pm$0.048	0.58$\pm$0.039	9.92	9.03
		w/o SA	0.8343 $\pm$ 0.0058	0.8787 $\pm$ 0.0046	238.72 $\pm$ 21.36	178.41 $\pm$ 17.12	0.93 $\pm$ 0.106	1.17 $\pm$ 0.134	12.72	10.82
		w/o RL	0.8414 $\pm$ 0.0049	0.8841 $\pm$ 0.0053	229.16 $\pm$ 19.62	174.63 $\pm$ 15.83	0.91 $\pm$ 0.098	1.15 $\pm$ 0.127	12.06	10.32
		w/o SA+RL	0.8273 $\pm$ 0.0065	0.8706 $\pm$ 0.0057	258.84 $\pm$ 23.72	193.77 $\pm$ 19.24	1.00 $\pm$ 0.116	1.25 $\pm$ 0.143	13.33	11.39
Banking77	Distilbert (Target F1: 0.8) (Interval τ : 10 rounds)	FFT	0.9420 $\pm$ 0.0010	0.9470 $\pm$ 0.0030	61.00 $\pm$ 1.63	79.33 $\pm$ 2.49	0.334 $\pm$ 0.008	0.445 $\pm$ 0.009	98.04	127.51
		LoRA	0.8774 $\pm$ 0.0060	0.8188 $\pm$ 0.0040	803.66 $\pm$ 38.38	933.00 $\pm$ 36.12	3.789 $\pm$ 0.230	4.457 $\pm$ 0.267	52.08	60.46
		PiSSA	0.9362 $\pm$ 0.0030	0.9228 $\pm$ 0.0160	393.60 $\pm$ 37.54	364.60 $\pm$ 17.13	1.847 $\pm$ 0.083	1.726 $\pm$ 0.007	25.51	23.62
		MiLoRA	0.9048 $\pm$ 0.0010	0.8822 $\pm$ 0.0060	627.33 $\pm$ 10.33	709.66 $\pm$ 16.21	2.890 $\pm$ 0.062	3.330 $\pm$ 0.061	40.65	45.98
		FlexLoRA	0.9176 $\pm$ 0.0038	0.9243 $\pm$ 0.0037	248.72 $\pm$ 19.41	268.63 $\pm$ 21.17	0.84 $\pm$ 0.078	0.91 $\pm$ 0.087	16.11	17.38
		LoRA-A2	0.9206 $\pm$ 0.0037	0.9274 $\pm$ 0.0036	233.84 $\pm$ 18.26	258.71 $\pm$ 20.36	0.80 $\pm$ 0.069	0.87 $\pm$ 0.078	15.22	16.68
		FedSVD	0.9244 $\pm$ 0.0028	0.9312 $\pm$ 0.0027	218.93 $\pm$ 17.12	243.67 $\pm$ 19.28	0.75 $\pm$ 0.058	0.83 $\pm$ 0.067	14.31	15.78
		pFedSpec	0.9443$\pm$0.0036	0.9709$\pm$0.0044	148.62$\pm$9.41	118.37$\pm$8.12	0.73$\pm$0.046	0.60$\pm$0.038	11.78	10.06
		w/o SA	0.9192 $\pm$ 0.0047	0.9388 $\pm$ 0.0038	308.41 $\pm$ 24.12	258.63 $\pm$ 21.08	0.97 $\pm$ 0.098	1.11 $\pm$ 0.118	18.51	16.72
		w/o RL	0.9271 $\pm$ 0.0039	0.9449 $\pm$ 0.0037	293.27 $\pm$ 23.05	248.72 $\pm$ 20.14	0.93 $\pm$ 0.089	1.07 $\pm$ 0.108	17.32	15.81
		w/o SA + RL	0.9142 $\pm$ 0.0056	0.9367 $\pm$ 0.0048	328.63 $\pm$ 26.41	278.12 $\pm$ 22.36	1.03 $\pm$ 0.108	1.17 $\pm$ 0.128	19.71	18.02
Banking77	Distilbert (Target F1: 0.8) (Interval τ : 100 rounds)	FFT	0.9460 $\pm$ 0.0010	0.9455 $\pm$ 0.0010	76.00 $\pm$ 7.25	94.00 $\pm$ 8.10	0.419 $\pm$ 0.031	0.510 $\pm$ 0.040	101.79	133.40
		LoRA	0.8660 $\pm$ 0.0070	0.8230 $\pm$ 0.0030	860.00 $\pm$ 55.87	980.00 $\pm$ 60.00	3.704 $\pm$ 0.278	4.920 $\pm$ 0.320	52.22	63.04
		PiSSA	0.9305 $\pm$ 0.0020	0.9295 $\pm$ 0.0080	410.00 $\pm$ 40.00	390.00 $\pm$ 35.00	1.900 $\pm$ 0.169	1.820 $\pm$ 0.051	26.22	25.30
		MiLoRA	0.9114 $\pm$ 0.0040	0.8767 $\pm$ 0.0080	650.00 $\pm$ 40.00	780.00 $\pm$ 45.00	2.820 $\pm$ 0.150	3.750 $\pm$ 0.420	41.80	50.40
		FlexLoRA	0.9196 $\pm$ 0.0038	0.9272 $\pm$ 0.0037	258.73 $\pm$ 19.42	283.64 $\pm$ 21.36	0.89 $\pm$ 0.078	0.97 $\pm$ 0.087	16.42	17.98
		LoRA-A2	0.9225 $\pm$ 0.0037	0.9303 $\pm$ 0.0036	243.81 $\pm$ 18.31	268.52 $\pm$ 20.18	0.85 $\pm$ 0.069	0.92 $\pm$ 0.078	15.71	17.08
		FedSVD	0.9264 $\pm$ 0.0028	0.9342 $\pm$ 0.0027	228.94 $\pm$ 17.12	253.71 $\pm$ 19.24	0.80 $\pm$ 0.058	0.88 $\pm$ 0.067	14.82	15.99
		pFedSpec	0.9471$\pm$0.0038	0.9746$\pm$0.0046	173.84$\pm$9.72	144.21$\pm$8.63	0.83$\pm$0.048	0.71$\pm$0.039	13.46	12.21
		w/o SA	0.9213 $\pm$ 0.0047	0.9398 $\pm$ 0.0038	318.42 $\pm$ 24.91	278.63 $\pm$ 21.34	1.01 $\pm$ 0.098	1.17 $\pm$ 0.118	19.41	17.81
		w/o RL	0.9282 $\pm$ 0.0039	0.9461 $\pm$ 0.0037	303.26 $\pm$ 23.42	263.71 $\pm$ 20.41	0.97 $\pm$ 0.089	1.11 $\pm$ 0.108	18.22	16.81
		w/o SA + RL	0.9153 $\pm$ 0.0056	0.9374 $\pm$ 0.0048	338.71 $\pm$ 26.83	293.84 $\pm$ 22.61	1.07 $\pm$ 0.108	1.21 $\pm$ 0.128	20.61	18.98

classes generally exhibit higher KL_{norm} , indicating larger divergence from the global distribution, while clients with richer data exhibit lower divergence. This confirms that heterogeneity is not only in sample size but also in distributional complexity. Initial ranks are randomly diversified across clients to avoid bias toward a fixed capacity regime and to enable exploration of different spectral representations before adaptation.

Figure 3 further illustrates how these heterogeneous clients evolve their ranks over time. Despite diverse initializations, client ranks progressively adapt toward stable mid-range values (approximately 16–64) in both datasets. This convergence suggests that extreme capacities (very low or very high ranks) are suboptimal under time-varying and heterogeneous data, and that the Q-learning policy naturally redistributes capacity toward a balanced operating region.

To explain this behavior, Figure 4 visualizes the learned Q-table that governs rank transitions. Each client maintains its own Q-table over discrete rank states and rank adjustment actions. Rows correspond to current ranks, and columns correspond to possible rank updates. High Q-values (red) indicate preferred rank transitions, while low values (blue) represent discouraged actions, with invalid transitions masked in gray.

The structure of the Q-table directly explains the observed dynamics. The learned policy consistently assigns higher Q-values to transitions toward mid-range ranks (e.g., 16, 32, 64), while penalizing both extreme ranks and large abrupt changes. This induces a stable exploration–exploitation trade-off, preventing oscillation and gradually guiding clients toward a balanced capacity regime.

Table 8Number of samples, classes, and KL_{norm} for 20News and Banking77 clients under $\alpha = 0.01$.

Client (FedAvg)	20News				Banking77			
	# Samples	# Classes	KL_{norm}	Init rank	# Samples	# Classes	KL_{norm}	Init rank
Client 0	1741	4	0.3603	4	1047	10	0.4209	4
Client 1	128	2	0.9177	4	316	6	1.0000	4
Client 2	1652	3	0.4917	64	977	8	0.5582	64
Client 3	2413	6	0.0000	2	754	11	0.5510	2
Client 4	32	2	1.0000	8	798	10	0.5452	8
Client 5	1442	4	0.5238	768	1669	17	0.0000	768
Client 6	1068	5	0.8457	512	968	16	0.3717	512
Client 7	1041	3	0.8714	64	1177	19	0.2930	64
Client 8	560	6	0.8337	2	872	23	0.4007	2
Client 9	1237	7	0.6085	1	1425	28	0.1669	1

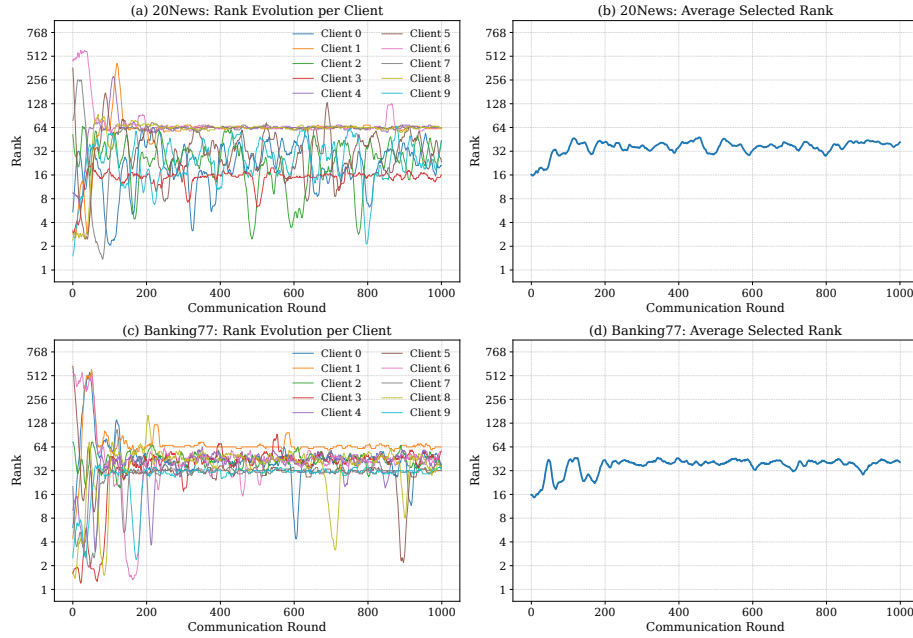


Figure 3: Dynamic rank adaptation behavior of pFedSpec. Top-left: per-client rank evolution on 20News. Top-right: average selected rank on 20News. Bottom-left: per-client rank evolution on Banking77. Bottom-right: average selected rank on Banking77. The results show that clients progressively converge toward mid-range ranks (e.g., 16–64), indicating stable capacity allocation driven by Q-learning.

5.2.6. Ablation Study on Top- $\mathcal{K}$ Spectrum Selection

We evaluate the effectiveness of the proposed Top- $\mathcal{K}$ spectrum selection under a highly heterogeneous setting with Dirichlet partitioning ($\alpha = 0.01$). The comparison is conducted between the proposed Top- $\mathcal{K}$ strategy and a Top-1 variant, where only the single highest-weight spectrum block is selected while all other components of pFedSpec remain unchanged. As shown in Table 9, Top- $\mathcal{K}$ consistently outperforms Top-1 across all four datasets and three backbone architectures. In particular, relative F1-score improvements of 1.23%, 0.85%, 2.04%, and 0.42% are achieved on 20News, Banking77, CIFAR-100, and UPMC-Food101, respectively, yielding an average improvement of 1.14%.

Importantly, these performance gains do not require a higher adapter rank or additional communication overhead. With Top- $\mathcal{K}$, the selected spectrum blocks are individually used to construct candidate adapters, which are

then combined through weighted averaging to form a single personalized adapter with the target rank. Thus, although Top- $\mathcal{K}$ considers multiple spectral blocks, the final adapter maintains the same rank and parameter size as Top-1. The consistent improvement across text, vision, and multimodal datasets demonstrates that multiple spectral blocks contain complementary information that is beneficial for personalization, whereas relying only on the single highest-weight block may discard useful spectral information. This ablation therefore supports the use of Top- $\mathcal{K}$ spectrum aggregation as an effective way to improve personalized adaptation without increasing the parameter or communication budget.

6. Limitations and Discussion

Although pFedSpec demonstrates consistent improvements in accuracy, convergence speed, and communication

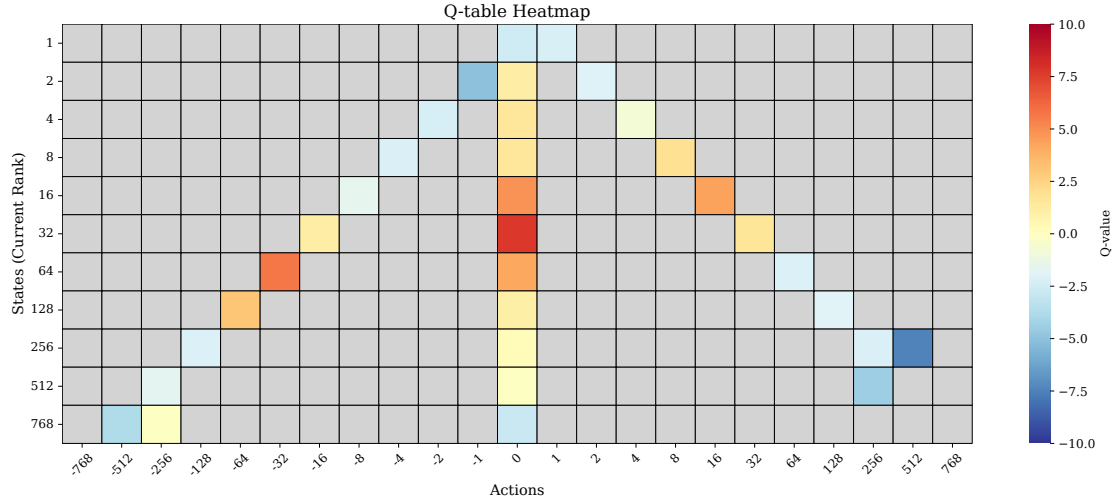


Figure 4: Visualization of the learned Q-table for rank adaptation. Each row corresponds to a current rank (state), and each column corresponds to a rank adjustment action. Valid transitions are colored according to their Q-values (blue: low, red: high), while invalid actions are masked in gray. The structure shows a clear preference for transitions toward mid-range ranks (e.g., 16, 32, 64), while large jumps and extreme ranks are penalized, reflecting the stability-efficiency trade-off learned by the policy.

Table 9

Ablation study of Top- $\mathcal{K}$ spectrum selection. Top- $\mathcal{K}$ consistently improves the final F1-score over the Top-1 variant while maintaining the same communication cost and adapter rank.

Dataset	Backbone	Top-1	Top- $\mathcal{K}$	Improvement
20News	DistilBERT	0.8940	0.9050	+1.23%
Banking77	DistilBERT	0.9638	0.9720	+0.85%
CIFAR-100	ResNet18	0.5890	0.6010	+2.04%
UPMC-Food101	CLIP	0.9510	0.9550	+0.42%
Average Improvement	—	—	—	+1.14%

efficiency across language, vision, and multimodal tasks, several limitations remain.

First, despite reducing communication costs, pFedSpec introduces extra local computation due to singular value decomposition and spectrum reallocation. Although these operations are performed infrequently and their cost is relatively small compared with full fine-tuning, they may still impose additional burdens on resource-constrained edge devices. Therefore, the overall system efficiency depends not only on communication savings but also on the computational capabilities of participating clients.

Second, the current method determines candidate windows based on spectral weight and learns client-specific ranks through reinforcement learning. While this mechanism effectively balances personalization and communication efficiency, it operates independently across layers and does not explicitly consider cross-layer interactions. Consequently, the allocation process may overlook dependencies between spectral components belonging to different layers, potentially leading to suboptimal resource utilization. Incorporating global layer-wise coordination or hierarchical spectrum allocation could further improve the expressiveness of the learned representations.

Third, the reinforcement learning framework introduces additional hyperparameters and training complexity. The quality of the learned policy depends on reward design and exploration strategies, and convergence may become unstable in extremely heterogeneous environments. Although our experiments show that rank learning consistently improves performance, the current implementation employs relatively simple state representations and action spaces. More sophisticated policy optimization techniques or meta-learning approaches may provide stronger adaptability and faster convergence.

Finally, although pFedSpec is evaluated on diverse datasets and model architectures, the experiments mainly focus on moderately sized language and vision models. The scalability of the proposed framework to very large foundation models with billions of parameters and thousands of participating clients remains to be investigated. In such settings, issues such as asynchronous communication, client dropout, and dynamic participation may interact with spectrum allocation mechanisms in nontrivial ways. Exploring these scenarios constitutes an important direction for future research.

7. Conclusion

This paper presented **pFedSpec**, a personalized federated fine-tuning framework that leverages spectral decomposition to enable adaptive parameter sharing under heterogeneous client distributions. Unlike existing PEFT-based FL approaches that employ a fixed low-rank structure for all clients, pFedSpec decomposes LoRA adapters into spectral components and dynamically allocates task-relevant spectral subspaces according to each client’s intrinsic rank characteristics. Furthermore, a reinforcement learning mechanism

is introduced to optimize spectrum selection and balance personalization with communication efficiency.

Comprehensive experiments across text, vision, multi-modal, and time-varying scenarios demonstrate that pFedSpec consistently achieves superior performance over state-of-the-art baselines. The proposed method not only improves predictive accuracy and convergence speed, but also substantially reduces communication overhead, highlighting the effectiveness of spectrum-aware personalization in heterogeneous federated environments. Ablation studies further verify the importance of both the spectrum allocation strategy and reinforcement learning policy in achieving these gains.

Overall, the results suggest that spectral representations provide a principled and scalable foundation for personalized federated fine-tuning. By exploiting the intrinsic low-rank structure of model updates, pFedSpec offers an effective pathway toward communication-efficient and adaptive FL for large foundation models. We hope that this work will stimulate further research on spectrum-based parameterization and adaptive resource allocation for next-generation personalized federated systems.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The datasets used in this study are all open source, and their sources have been duly cited in the paper.

References

- Achiam, J., Adler, S., Agarwal, S., Ahmad, L., Akkaya, I., Aleman, F.L., Almeida, D., Altenschmidt, J., Altman, S., Anadkat, S., et al., 2023. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774* <https://doi.org/10.48550/arXiv.2303.08774>.
- Adinugroho, S., Wihandika, R.C., Adikara, P.P., 2022. Newsgroup topic extraction using term-cluster weighting and pillar k-means clustering. *International journal of computers and applications* 44, 357–364. <https://doi.org/10.1080/1206212X.2020.1757246>.
- Alayrac, J.B., Donahue, J., Luc, P., Miech, A., Barr, I., Hasson, Y., Lenc, K., Mensch, A., Millican, K., Reynolds, M., et al., 2022. Flamingo: a visual language model for few-shot learning. *Advances in neural information processing systems* 35, 23716–23736. <https://doi.org/10.52202/068431-1723>.
- Arisdakessian, S., Wehbi, O., Wahab, O.A., Mourad, A., Otrók, H., Guizani, M., 2025. A two-level dirichlet framework for heterogeneous federated network. *IEEE Transactions on Network Science and Engineering* <https://doi.org/10.1109/TNSE.2025.3597541>.
- Bai, J., Chen, D., Qian, B., Yao, L., Li, Y., 2024. Federated fine-tuning of large language models under heterogeneous tasks and client resources. *Advances in Neural Information Processing Systems* 37, 14457–14483. <https://doi.org/10.52202/079017-0461>.
- Bossard, L., Guillaumin, M., Van Gool, L., 2014. Food-101—mining discriminative components with random forests. *European conference on computer vision*, 446–461 https://link.springer.com/chapter/10.1007/978-3-319-10599-4_29.
- Cho, Y.J., Liu, L., Xu, Z., Fahrezi, A., Joshi, G., 2024. Heterogeneous lora for federated fine-tuning of on-device foundation models. *Proceedings of the 2024 conference on empirical methods in natural language processing*, 12903–12913 <https://doi.org/10.18653/v1/2024.emnlp-main.717>.
- Chowdhery, A., Narang, S., Devlin, J., Bosma, M., Mishra, G., Roberts, A., Barham, P., Chung, H.W., Sutton, C., Gehrmann, S., et al., 2023. Palm: Scaling language modeling with pathways. *Journal of machine learning research* 24, 1–113. <http://jmlr.org/papers/v24/22-1144.html>.
- Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., et al., 2021. An image is worth 16x16 words: Transformers for image recognition at scale. *International Conference on Learning Representations (ICLR)* <https://doi.org/10.48550/arXiv.2010.11929>.
- Gama, J., Žliobaitė, I., Bifet, A., Pechenizkiy, M., Bouchachia, A., 2014. A survey on concept drift adaptation. *ACM computing surveys (CSUR)* 46, 1–37. <https://doi.org/10.1145/2523813>.
- Goldberger, Gordon, Greenspan, 2003. An efficient image similarity measure based on approximations of kl-divergence between two gaussian mixtures. *Proceedings Ninth IEEE International conference on computer vision*, 487–493 <https://doi.org/10.1109/ICCV.2003.1238387>.
- Guo, P., Zeng, S., Wang, Y., Fan, H., Wang, F., Qu, L., 2025. Selective aggregation for low-rank adaptation in federated learning. *International Conference on Learning Representations 2025*, 99003–99027. <https://doi.org/10.48550/arXiv.2410.01463>.
- Han, Z., Gao, C., Liu, J., Zhang, J., Zhang, S.Q., 2024. Parameter-efficient fine-tuning for large models: A comprehensive survey. *arXiv preprint arXiv:2403.14608* <https://doi.org/10.48550/arXiv.2403.14608>.
- Hao, J., Wu, Y., Payani, A., Lee, M., Liu, M., 2025. Personalized federated fine-tuning for heterogeneous data: An automatic rank learning approach via two-level lora. *arXiv preprint arXiv:2503.03920*.
- Hoffmann, J., Borgeaud, S., Mensch, A., Buchatskaya, E., Cai, T., Rutherford, E., Casas, D., Hendricks, L.A., Welbl, J., Clark, A., et al., 2022. Training compute-optimal large language models. *arXiv preprint arXiv:2203.15556* <https://doi.org/10.48550/arXiv.2203.15556>.
- Houlsby, N., Giurgiu, A., Jastrzebski, S., Morrone, B., De Laroussilhe, Q., Gesmundo, A., Attariyan, M., Gelly, S., 2019. Parameter-efficient transfer learning for nlp. *International conference on machine learning*, 2790–2799 <https://api.semanticscholar.org/CorpusID:59599816>.
- Hu, J.E., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Chen, W., 2021. Lora: Low-rank adaptation of large language models. *International Conference on Learning Representations* <https://api.semanticscholar.org/CorpusID:235458009>.
- Koo, J., Jang, M., Ok, J., 2025. Towards robust and efficient federated low-rank adaptation with heterogeneous clients. *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 416–429 <https://doi.org/10.18653/v1/2025.acl-long.19>.
- Lee, S., Park, S., Lee, D.B., Wagner, D., Seong, H., Bocklet, T., Lee, J., Hwang, S.J., 2026. Fedsvd: Adaptive orthogonalization for private federated learning with lora. *Advances in Neural Information Processing Systems* 38, 119733–119757. <https://doi.org/10.52202/085713-3998>.
- Li, T., Sahu, A.K., Zaheer, M., Sanjabi, M., Talwalkar, A., Smith, V., 2020. Federated optimization in heterogeneous networks. *Proceedings of Machine learning and systems* 2, 429–450. <https://api.semanticscholar.org/CorpusID:59316566>.
- Liu, H., Brailsford, T., Bull, L., 2024. Resnet18 performance: Impact of network depth and image resolution on image classification. *Proceedings of the 2024 8th international conference on advances in artificial intelligence*, 351–356 <https://doi.org/10.1145/3704137.3704173>.
- Loukas, L., Stogiannidis, I., Diamantopoulos, O., Malakasiotis, P., Vassos, S., 2023. Making llms worth every penny: Resource-limited text classification in banking. *Proceedings of the fourth ACM international conference on AI in finance*, 392–400 <https://doi.org/10.1145/3604237.3626891>.
- Lu, J., Liu, A., Dong, F., Gu, F., Gama, J., Zhang, G., 2018. Learning under concept drift: A review. *IEEE transactions on knowledge and data engineering* 31, 2346–2363. <https://doi.org/10.1109/TKDE.2018.2876857>.
- McMahan, B., Moore, E., Ramage, D., Hampson, S., y Arcas, B.A., 2017. Communication-efficient learning of deep networks from decentralized

- data. Artificial intelligence and statistics , 1273–1282<https://api.semanticscholar.org/CorpusID:14955348>.
- Meng, F., Wang, Z., Zhang, M., 2024. Pissa: Principal singular values and singular vectors adaptation of large language models. *Advances in Neural Information Processing Systems* 37, 121038–121072. <https://doi.org/10.52202/079017-3846>.
- Patil, M.K., Patil, S., BR, S.S., Mythreyi, S., AM, A.M., 2026. Fs-dpo-qwen: Online courses review-based sentiment analysis using qwen 1.5-7b with few-shot prompting and direct preference optimization. 2026 2nd International Conference on Intelligent Systems and Computational Networks (ICISCN) , 1–9<https://doi.org/10.1109/ICISCN67954.2026.11566090>.
- Sanh, V., Debut, L., Chaumond, J., Wolf, T., 2019. Distilbert, a distilled version of bert: smaller, faster, cheaper and lighter. *arXiv preprint arXiv:1910.01108* <https://doi.org/10.48550/arXiv.1910.01108>.
- Shen, S., Li, L.H., Tan, H., Bansal, M., Rohrbach, A., Chang, K.W., Yao, Z., Keutzer, K., 2021. How much can clip benefit vision-and-language tasks? *arXiv preprint arXiv:2107.06383* <https://doi.org/10.48550/arXiv.2107.06383>.
- Sun, Y., Li, Z., Li, Y., Ding, B., 2024. Improving lora in privacy-preserving federated learning 2024, 17978–17994. <https://doi.org/10.48550/arXiv.2403.12313>.
- Team, G., Anil, R., Borgeaud, S., Alayrac, J.B., Yu, J., Soricut, R., Schalkwyk, J., Dai, A.M., Hauth, A., Millican, K., et al., 2023. Gemini: a family of highly capable multimodal models. *arXiv preprint arXiv:2312.11805* <https://doi.org/10.48550/arXiv.2312.11805>.
- Wang, H., Li, Y., Wang, S., Chen, G., Chen, Y., 2025. Milora: Harnessing minor singular components for parameter-efficient llm finetuning. *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)* , 4823–4836<https://doi.org/10.18653/v1/2025.naacl-long.248>.
- Yang, L., Zhang, W.E., Nguyen, N.K., Hu, P., Shu, Y., Chen, W., Sim, M.Y., 2025. Fedilora: Heterogeneous lora for federated multimodal fine-tuning under missing modalities. *arXiv preprint arXiv:2509.06984* https://ui.adsabs.harvard.edu/link_gateway/2025arXiv250906984Y/doi:10.48550/arXiv.2509.06984.
- Yang, Y., Long, G., Shen, T., Jiang, J., Blumenstein, M., 2024. Dual-personalizing adapter for federated foundation models. *Advances in Neural Information Processing Systems* 37, 39409–39433. <https://doi.org/10.52202/079017-1245>.
- Yun, S., Chae, S., Lee, D., Ro, Y., 2025. Soma: Singular value decomposed minor components adaptation for domain generalizable representation learning. *Proceedings of the Computer Vision and Pattern Recognition Conference* , 25602–25612<https://doi.org/10.1109/CVPR52734.2025.02384>.
- Zhang, Q., Chen, M., Bukharin, A., Karampatziakis, N., He, P., Cheng, Y., Chen, W., Zhao, T., 2023. Adalora: Adaptive budget allocation for parameter-efficient fine-tuning. *arXiv preprint arXiv:2303.10512* <https://doi.org/10.48550/arXiv.2303.10512>.
- Zheng, Y., Huang, H., Chen, J., 2024. Comparative analysis of various models for image classification on cifar-100 dataset. *Journal of Physics: Conference Series* 2711, 012015. <https://doi.org/10.1088/1742-6596/2711/1/012015>.